

Multi-Representational Reference Discovery for Progressive Retrieval Attention: Semantic, Lexical, and Iterative Routing Across Four Evidence Tasks

Jorge Simao

EInnovator R&D – Center for the Sciences of Computation, Cognition, and Complexity

August 23, 2026

Abstract

Progressive Retrieval Attention (PRA) separates a large logical memory from the smaller native-key/value working set admitted to attention. We study how PRA discovers that set using contextual semantic gists, exact and approximate model-token matching, BM25, token n -grams, and hybrid traversal. In the original four-dataset Qwen study, root and successor preferences differ on three datasets, while static score and rank fusion do not beat the best fixed channels. We then implement posting-list candidate narrowing and reproduce exhaustive top-four results for every paired query under both Qwen and SmolLM tokenizers. At 4,096 chunks, exact lookup is 283–300 $\times$ faster than exhaustive Python scoring; fuzzy cascades are 56–64 $\times$ faster but remain too slow for serving. A fresh Gate B confirmation uses 20 validation and 50 held-out identities per dataset, two frozen model families, four chunks/128 native-K/V tokens, five selector seeds, and complete causal controls. Channel geometry replicates: BM25 leads Hotpot recall (0.665/0.662), while approximate matching leads QASPER (0.747/0.783). Hotpot memory yields resolved gold-answer likelihood gains for both models, and the oracle beats irrelevant memory on both datasets and models. QASPER fixed-channel gains over disabled memory remain unresolved, and generated EM/F1 remains weak. Observable selection and referential calibration do not close these gaps. The supported result is therefore an indexed, portable discovery mechanism with causal native-K/V evidence on Hotpot, not a general QA improvement.

1 Introduction

PRA treats long context as addressable logical memory rather than a token sequence that must remain fully active [12, 13]. The system must find a tiny conceptual working set before a separate materializer decides which native K/V can participate in attention. Its native-KV design therefore separates stable object identity, compact routing representations, and detailed layer-specific state. This raises a distinct question: *what evidence should discover a candidate memory object?*

Current routing can score a contextual query $q_{t,\ell}$ against chunk gist $g_{i,\ell}$:

$$s_{i,\ell}^{sem} = \cos(q_{t,\ell}, g_{i,\ell}).$$

This captures contextual relatedness but need not preserve exact sequence identity. PRA, however, already has access to the host model tokenizer. Exact token spans, BM25-weighted terms, ordered token subsequences, and approximate subword matches recover evidence that semantic

routing misses. They do not reliably resolve paraphrases, implicit relations, or ambiguous names, so contextual gists remain necessary. The four datasets below do not yield one winning formula, and static fusion does not dominate. More importantly, the best representation often changes after a root chunk enters the state and exposes a successor address. Retrieval representation is therefore a state-dependent control decision, with separate root action S_{root} and successor action S_{succ} .

Paper 2 establishes the frozen-backbone native-K/V integration and identifies routing as the bottleneck [14]. Paper 2.5 studies bounded associative discovery and exposes entry-routing, topology, granularity, and interference limits [10]. This paper asks which representations should generate and rank root and successor candidates. Paper 3 separately asks which discovered candidates, and how much detail, should be materialized [11]. Keeping discovery and physical participation separate is central to the experimental design.

2 Hybrid Reference Discovery

At hop h , let

$$\mathcal{C}_h = \mathcal{C}_h^{explicit} \cup \mathcal{C}_h^{tok-exact} \cup \mathcal{C}_h^{tok-approx} \cup \mathcal{C}_h^{semantic} \cup \mathcal{C}_h^{assoc}.$$

Candidates retain URI/chunk identity, layer, raw channel scores, discovery hop, parent, provenance, rank, confidence, and an explicit calibration-status flag. Exact handles may resolve deterministically; approximate channels generate candidates.

An application can also pass stable reference handles with a generation request. The model validates those URIs against its registered reference cache and forwards the resolved URI set to every routing hop. Matching rows receive deterministic priority with provenance `explicit_reference`; unknown handles fail before generation. This is product wiring rather than an oracle: the caller must possess the URI, and no evidence annotation enters the resolver.

The implementation compares a progressive resolution cascade, candidate union, score fusion, token-native scoring followed by semantic reranking, semantic scoring followed by token-native reranking, and iterative hybrid traversal. Classical BM25 over decoded word terms is retained as a strong baseline rather than absorbed into the proposed mechanism.

3 Token-Native Reference Matching

The proposed non-neural resolver operates over the host model tokenizer. For each memory object we retain raw token IDs, normalized token spans, decoded subword strings, and title/name/alias boundaries. This permits exact identity matching without a second NLP vocabulary, while still allowing controlled normalization of tokenizer-specific leading-space, punctuation, and case artifacts.

A progressive cascade is evaluated:

explicit $\rightarrow$ exact token span $\rightarrow$ weighted/ordered token $\rightarrow$ approximate token $\rightarrow$ semantic gist.

Earlier high-confidence stages may resolve or sharply narrow candidates before later stages run.

For weighted overlap, let Q and R_i denote query and reference token multisets or aligned spans, and let $w(t)$ be a corpus-dependent token weight. A simple form is

$$s_i^{overlap} = \frac{\sum_{t \in \text{uniq}(Q) \cap \text{uniq}(R_i)} w(t)}{\sum_{t \in \text{uniq}(R_i)} w(t)}.$$

The present token-native score uses token-space inverse document frequency,

$$w(t) = \log \frac{N + 1}{df(t) + 1},$$

which can automatically downweight domain-generic tokens. Ordered subsequence overlap preserves multi-token identity while tolerating intervening material. A fixed stop list can instead remove common function words before retrieval. Token n -gram posting lists preserve local order without requiring an exact full-span match.

Approximate matching is attempted only after exact or high-confidence matching fails or remains ambiguous. Character-trigram postings first narrow the candidate set; sequence similarity is then evaluated only in that pool. Two additional diagnostic channels use bounded token edit distance over short subspans and cosine similarity between mean host-model input embeddings. The latter reuses the model’s own embedding matrix rather than adding an independently trained representation. These channels are available for measurement, but are not silently folded into the product default.

3.1 Indexed Candidate Generation

Exhaustive scoring obscures the systems question because every query pays for every logical chunk. The production sidecar therefore stores postings for raw token IDs, normalized pieces, token n -grams, BM25 terms, character trigrams, and explicit reference URIs. A query forms the deduplicated union

$$\mathcal{U}(q) = \mathcal{P}_{tok}(q) \cup \mathcal{P}_{ngram}(q) \cup \mathcal{P}_{bm25}(q) \cup \mathcal{P}_{char}(q) \cup \mathcal{P}_{uri}(q),$$

then applies the requested channel only to rows in $\mathcal{U}(q)$, capped by a fixed candidate-pool budget. Semantic routing remains a tensor operation over gist states; indexed lexical retrieval narrows the token-native work. Channel unions deduplicate stable row identities before ranking.

Index extension is immutable: adding references rebuilds corpus document frequencies and row-aligned postings and returns a new sidecar. This costs more than an in-place append, but prevents stale IDF statistics and lets concurrent readers retain a consistent snapshot. We consequently report construction, warm-query, memory, and update costs separately.

Classical word-level BM25 remains a strong baseline [8]. Approximate string matching is likewise a mature field [6]. The scientific question is whether these representations complement native semantic gists inside PRA’s stateful discovery process, not whether conventional information retrieval is unnecessary.

3.2 Confidence and Provenance

A candidate record includes raw channel scores and an explicit confidence field with stable intended semantics:

$$c_i \approx P(r_i \text{ is a valid referent} \mid q, \mathbf{s}_i, p_i),$$

where $\mathbf{s}_i$ contains channel-specific evidence and p_i records discovery provenance. The same numerical score from an exact token match, an approximate token match, and a semantic gist is not assumed to have the same reliability.

Raw discovery scores are marked uncalibrated in runtime traces. We fit calibration only on validation candidates and evaluate Brier score, expected calibration error, reliability, and selective accuracy/coverage. This makes the discovery interface testable for subsequent control:

$$c_i > \tau_{act} \Rightarrow \text{materialize},$$

$$\tau_{search} < c_i \leq \tau_{act} \Rightarrow \text{retrieve/disambiguate further},$$

with low-confidence cases eligible for abstention. External side-effecting action remains outside the scope of this paper, but the same confidence semantics can later support act/search/ask/abstain policies in agent systems.

4 Multi-Hop Reference Following

A useful pattern is

$$Q \xrightarrow{\text{semantic}} C_1 \xrightarrow{\text{token reference}} C_2 \xrightarrow{\text{reasoning}} A.$$

After selecting C_1 , its indexed token sequence becomes the lexical query for the next bounded hop. Its semantic gist remains the semantic frontier. Both channels therefore score the same aligned candidate rows before the existing branch, beam, depth, and unique-chunk controls admit C_2 . This produces a dynamically traversed chunk–reference graph without requiring a complete symbolic knowledge graph in advance.

If reference structure is informative, sparse traversal may replace broad semantic expansion:

same evidence recall with fewer candidates/materialized tokens.

5 Implementation

The implementation is intentionally split into a discovery sidecar and the inherited Paper 2.5 traversal. `TokenNativeIndex` in `src/prahf/hybrid_discovery.py` begins at line 298; `from_gist_index` at line 391 tokenizes each source once and slices it with the cache’s stable token offsets. Its rows must match `GistIndex.records` exactly; alignment is checked at line 458 before routing. `DiscoveryCandidate` (line 218) retains URI, chunk and layer identity, all raw channel scores, spans, hop, parent, provenance, rank, and confidence. Longest exact spans, token n -grams, bounded edit similarity, and automatic aliases begin at lines 80, 115, 125, and 173.

The sparse path is visible rather than hidden behind a generic search call. `_candidate_indices` at lines 516–580 accumulates posting votes, reserves part of the bounded pool for the highest semantic gist rows, adds any explicit URI rows, and deduplicates their union. The payload-byte estimate and immutable rebuild are at lines 583 and 611. `score` begins at line 630; lines 668–680 gate expensive feature families so a sparse exact query does not compute edit, BM25, embedding, and sequence scores it will never use. Channel-specific decisions are made at lines 848–886, where an explicit URI overrides learned or heuristic scores while retaining its provenance.

Crucially, this code does not add a second traversal engine. `IterativeGistRouter.route` in `src/prahf/iterative.py` begins at line 413. Optional token arguments are validated at lines 433–450. Root indexed scoring at lines 470–479 and successor scoring at lines 540–549 both request sparse row maps. The router scatters those rows back into its aligned tensor index, after which the original visited-set, per-parent branch, beam, confidence, and unique-chunk budget apply unchanged. An admitted chunk contributes its gist and token IDs to the next frontier. The retrieval graph counts index queries and expensive candidate comparisons separately.

The product path exposes `token_iterative` and `hybrid_iterative` configuration modes. In `src/prahf/model.py`, `PRAForCausalLM` constructs the aligned sidecar and passes explicit URI state at lines 440–458. After traversal stops, the existing identity mapper resolves selected chunks across consumption layers at lines 469–472; only then are graph nodes marked materialized at lines 477–481. The public `generate` API accepts reference handles at line 491, validates unknown URIs at lines 504–512, and records the resolved set in returned statistics at line 568. Thus selected conceptual chunks, requested chunks, and active physical K/V remain separately observable.

The channel-geometry boundary is independently reusable. `src/prahf/channel_geometry.py` defines identity-set precision/recall and the deterministic channel oracle at lines 36–55, rare new-address detection at lines 90–106, and the gold-rejecting observable selector beginning at line 123. The experiment adapter in `experiments/paper2_6_hybrid_pra/run_channel_geometry.py`

reconstructs one 32-token mean gist per frozen natural-feature span beginning at line 85, extracts root ranks, score gaps, lexicality, and evidence dispersion beginning at line 206, and starts the complete artifact-producing protocol at line 395. This split lets another implementation replace the diagnostic policy without changing candidate identity, traversal, or K/V materialization semantics.

The follow-up primitives are similarly explicit. Rank-only fusion, headroom decomposition, and the competitive-address predicate are at lines 63–122 of `src/prahf/channel_geometry.py`. The root/successor cross-product in `experiments/paper2_6_hybrid_pra/run_channel_selection.py` begins at line 216; the validation-only linear selector begins at line 386; extended collision controls begin at line 543; and the complete run begins at line 866. The versioned action contract is constructed at line 599 and emitted as `search_method_action_spec.json`.

Final diagnostics are reusable rather than embedded in the paper. Leakage validation and calibration summaries begin at lines 51 and 141 of `src/prahf/confidence_diagnostics.py`; paired channel bootstrap begins at line 221 and schema validation at line 303. The frozen candidate replay begins at line 221 of `experiments/paper2_6_hybrid_pra/run_final_iteration.py`; cohort stability, calibration, the consistency gate, and the UsefulAddress proxy begin at lines 397, 449, 488, and 574. The artifact-producing entry point begins at line 825. These boundaries let Paper 3.5 consume methods and confidence fields without importing Paper 2.6’s gold evaluation labels.

6 Research Questions and Success Levels

1. Does hybrid discovery improve supporting-evidence recall at matched active budget?
2. Which discovery signals dominate entry versus later hops?
3. Can iterative reference following outperform simply increasing semantic top- k ?
4. Do candidate provenance and confidence improve downstream materialization/control decisions?
5. When does token-level reference structure fail to add value over contextual gists?
6. Do retrieval improvements survive end-to-end answer generation?

We distinguish two success levels. A *discovery-level positive result* requires improved required-evidence recall at matched active cost, or matched recall with materially lower active memory/cost, supported statistically and by causal controls. An *end-to-end positive result* additionally requires a resolved improvement in downstream QA, such as paired EM/F1 improvement with an appropriate confidence interval or significance test on at least one natural multi-hop dataset.

This distinction is essential. A robust discovery Pareto improvement remains scientifically meaningful independently of whether a frozen consumer converts it into answer gains. Gate B tests that conversion separately; retrieval recall is never described as improved reasoning or QA performance.

7 Tokenization Robustness

Token-native matching introduces a specific invariance question: can the resolver recognize the same referent when its surface form induces a different tokenizer sequence? We therefore make tokenization robustness a first-class experiment rather than a minor tokenizer-family ablation.

For a fixed referent r , construct controlled variants

$$\mathcal{P}(r) = \{r_{\text{case}}, r_{\text{space}}, r_{\text{punct}}, r_{\text{hyphen}}, r_{\text{split}}, r_{\text{typo}}, r_{\text{abbr}}, r_{\text{alias}}, \dots\}.$$

The expanded controlled suite covers case, punctuation, leading-space artifacts, whitespace splitting, hyphenation, concatenation, one-character misspellings, abbreviations, aliases, full-width Unicode and mathematical-letter compatibility forms, near entities, shared prefixes, numeric identifiers, URL/domain overlap, same-name and same-class ambiguity, wrong relations, stale aliases, two plausible references, and contradictory high-overlap names. Normalization uses Unicode NFKC before case folding. Conservative aliases are extracted automatically from headings and parenthetical names; application-supplied aliases remain available when a stronger entity registry exists.

We measure degradation separately for exact-span, weighted/ordered, approximate-token, cascade, and iterative-hybrid conditions. The objective is an invariance curve showing how rapidly each mechanism loses referent accuracy as the surface form moves away from exact token identity.

8 Wrong-Reference Stress Testing

Average recall is insufficient if token-native matching becomes confidently wrong under ambiguity. We implement controlled conflicts in which an admitted chunk exposes a high-overlap distractor, a stale alias, an instance of the right class, the right entity under the wrong relation, or two lexically plausible referents. These are deterministic mechanism fixtures, not a claim that entity resolution is solved.

A particularly diagnostic condition is

$$s_{\text{token}}(A) \gg s_{\text{token}}(B), \quad s_{\text{semantic}}(B) \gg s_{\text{semantic}}(A),$$

with only one candidate being the valid referent. The controlled test measures wrong-candidate confidence and target recovery after the conflicting hop. Gate B adds naturally occurring top-choice errors, validation-only selective calibration, and wrong-memory generation controls, but it is not a curated natural same-name coreference corpus. A useful hybrid method should degrade gracefully rather than convert ambiguous token evidence into high-confidence catastrophic routing errors.

9 Experimental Protocol

9.1 Controlled Mechanism Tasks

We construct 20 deterministic chains per condition. A semantic entry identifies the first chunk, which exposes a token address for a target placed behind two stronger semantic distractors. Fifteen surface-form and referential conditions use the same two-chunk request budget and the same candidate layout. These are mechanism probes, not independent natural examples.

9.2 Natural Evidence Retrieval

We reuse the identity-disjoint Paper 2 feature protocol: frozen Qwen3-0.6B attention-input hidden states at layer 27, 32-token mean-gist chunks, revision `c1899de...e47ca`, and the question-exponential query. Validation contains eight HotpotQA and eight QASPER examples; held-out test contains 16 of each. No model weights are loaded in this study. B0–B4 and H1–H5 receive a final budget of four requested chunks. A1 deliberately buys breadth with eight semantic chunks and is not a matched-budget result. O1 admits annotated evidence first as a ceiling. No candidate is materialized.

The hybrid entry/late semantic weights are selected on validation from $\{0.4, 0.6, 0.8\} \times \{0.1, 0.3, 0.5\}$, with token weights as complements. Entry weights 0.4 and 0.6 tie in aggregate

validation recall; the semantic-entry/token-later tie-break selects (0.6,0.1). Test identities are not used for selection. Confidence uses a monotone ten-bin histogram fitted only on validation candidates.

The extension reuses Paper 2.5’s separately frozen natural-graph cohort: 24 validation and 24 test 2Wiki examples, plus 18 validation and 18 test MuSiQue examples. Labelled evidence spans map through the fast tokenizer before the routing partition is constructed. Frozen token hidden states are averaged over the same 32-token local spans, yielding one gist per candidate chunk; the question-exponential query and layer remain unchanged. Six primary channels—gist, exact, BM25, approximate token, static token-to-semantic reranking, and two-hop iterative hybrid—all request exactly four unique chunks. Iteration uses branch two per step, depth two, at most four unique chunks, and reports all comparisons. The (0.6,0.1) entry/later semantic weights are inherited without retuning. These cohorts remain below the requested 50 held-out units per dataset, so all new comparisons are explicitly cohort-bounded.

Per-example explanatory geometry includes query/evidence token overlap, IDF-weighted overlap, named-entity overlap, evidence regions, evidence extent and gaps, annotated depth, root ranks, score gaps, and channel disagreement. Gold-derived geometry is never passed to the adaptive baseline. Its fixed rule can see query rarity, root score gaps, channel disagreement, and whether an already admitted hop-one chunk exposes a rare address. The selector API rejects gold fields. Best fixed channels are selected on validation; the per-example oracle is evaluated only as an upper bound.

10 Baselines and Conditions

Condition	Discovery	Role
X0	explicit registered URI	deterministic product path
B0	gist-only PRA	semantic baseline
B1	BM25/word lexical	classical IR baseline
B2	exact token span/name	identity baseline
B3	weighted/ordered token	token-native baseline
B4	approximate token	fallback baseline
B5	token n -gram	order-preserving diagnostic
B6	bounded token edit	typo diagnostic
B7	host input embedding	tokenizer-native dense diagnostic
H1	semantic $\cup$ token-native	hybrid recall
H2	token-native $\rightarrow$ semantic	reranking
H3	semantic $\rightarrow$ token-native	reverse reranking
H4	token cascade + semantic fallback	cascade
H5	iterative token/semantic hybrid	main proposal
A1	broad semantic top- k	breadth baseline
O1	annotated/oracle	ceiling

Candidate budgets are matched for B0–B4 and H1–H5 in the original discovery experiment. B5–B7 enter the expanded diagnostic and interaction suites. X0 is exercised as an API/mechanism contract rather than scored as learned discovery. A1 and O1 are labeled ceilings; no materialization budget is consumed in the original discovery-only experiment.

11 Results

Table 1: Held-out evidence discovery. R, P, and M are supporting-evidence recall, precision, and MRR. Principal conditions request four chunks; A1 requests eight.

Condition	HotpotQA			QASPER		
	R	P	M	R	P	M
B0 gist	.151	.172	.276	.038	.062	.130
B1 BM25	.398	.453	.875	.042	.094	.135
B2 exact	.287	.328	.568	.178	.312	.443
B3 weighted	.287	.328	.568	.061	.109	.214
B4 approximate	.301	.359	.734	.037	.078	.141
H1 union	.209	.250	.365	.038	.062	.130
H2 token→semantic	.287	.344	.635	.044	.078	.125
H3 semantic→token	.279	.328	.635	.052	.094	.156
H4 cascade	.240	.281	.474	.126	.203	.307
H5 iterative hybrid	.279	.312	.646	.081	.125	.177
A1 broad semantic	.310	.180	.363	.088	.078	.155
O1 oracle	.876	1.000	1.000	.633	1.000	1.000

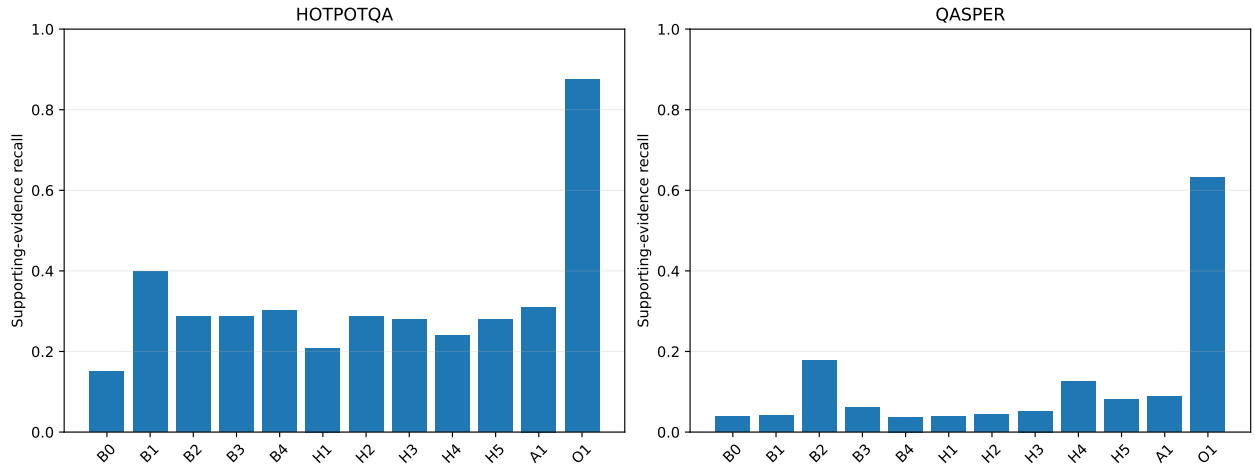


Figure 1: Held-out evidence recall. H5 improves gist-only recall on HotpotQA at the same four-chunk request budget, but does not surpass decoded-word BM25. QASPER favors exact spans. A1 uses twice the requested budget.

On HotpotQA, H5 improves recall over B0 by 0.129. A paired 10,000-sample bootstrap over the 16 test identities gives a 95% interval of [0.018, 0.242]; ten examples improve, four worsen, and two tie. B1 remains stronger, so the result supports heterogeneous discovery but not superiority of the present fusion rule. H5 reaches 90% of A1’s recall (0.279 versus 0.310) while requesting half as many chunks; this cross-budget comparison is descriptive.

On QASPER, H5 improves recall by 0.043, but its interval $[-0.014, 0.111]$ crosses zero (four improve, two worsen, ten tie). Exact spans instead improve over B0 by 0.140 with interval [0.053, 0.244]. This is evidence for regime dependence: QASPER’s scientific evidence in this small slice benefits from exact continuity, while repeated traversal adds little.

11.1 Four-Dataset Channel Geometry

Table 2 holds the request budget, chunk size, model, layer, and query representation fixed while changing only discovery channel. The winner changes across cohorts. QASPER favors exact spans; HotpotQA and 2Wiki favor decoded-word BM25; MuSiQue favors normalized approximate token matching, with static hybrid close behind. Static fusion improves over gist in every cohort but never beats the strongest single channel. The result is not “hybrid search wins.” It is that semantic and lexical representations fail differently, and their relative utility changes by example and task.

Table 2: Held-out recall/precision at four requested 32-token chunks. Bold marks the largest recall in each row; 95% paired-example bootstrap intervals for those winners appear below.

Dataset (n)	Gist	Exact	BM25	Approx.	Static hybrid	Iterative hybrid
QASPER (16)	.038/.062	.178 /.312	.042/.094	.037/.078	.044/.078	.081/.125
HotpotQA (16)	.151/.172	.287/.328	.398 /.453	.301/.359	.288/.344	.279/.312
2Wiki (24)	.132/.188	.196/.240	.262 /.312	.203/.250	.203/.250	.186/.250
MuSiQue (18)	.108/.333	.136/.417	.132/.431	.156 /.514	.151/.500	.146/.472

Best-channel recall intervals are QASPER [.096,.271], HotpotQA [.282,.516], 2Wiki [.176,.353], and MuSiQue [.122,.194].

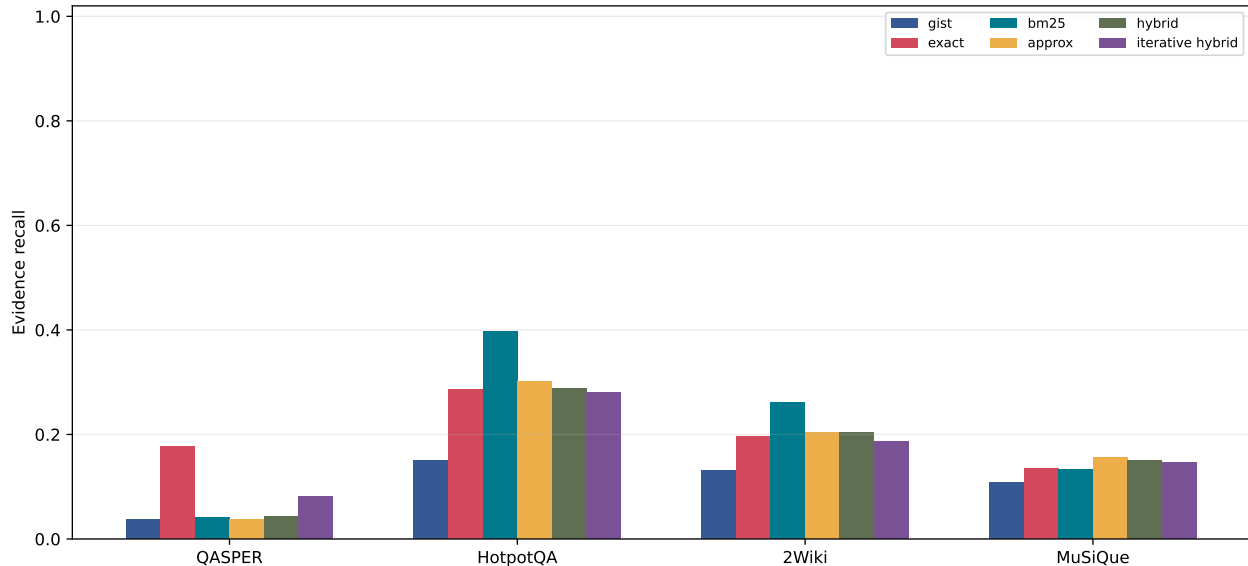


Figure 2: Matched-budget recall across retrieval channels and datasets. Different lexical channels win each cohort; neither static nor iterative hybrid dominates.

Precision moves with recall rather than revealing a hidden fusion win. The strongest precision is 0.312 for QASPER exact, 0.453 for Hotpot BM25, 0.312 for 2Wiki BM25, and 0.514 for MuSiQue approximate matching. Across all channel pairs, mean selected-set Jaccard is only 0.256 on QASPER, 0.325 on Hotpot, 0.369 on 2Wiki, and 0.326 on MuSiQue. In particular, gist–BM25 overlap is 0.045, 0.143, 0.152, and 0.119. The channels are not redundant, but naive fusion often spends its fixed budget on complementary distractors as well as complementary evidence.

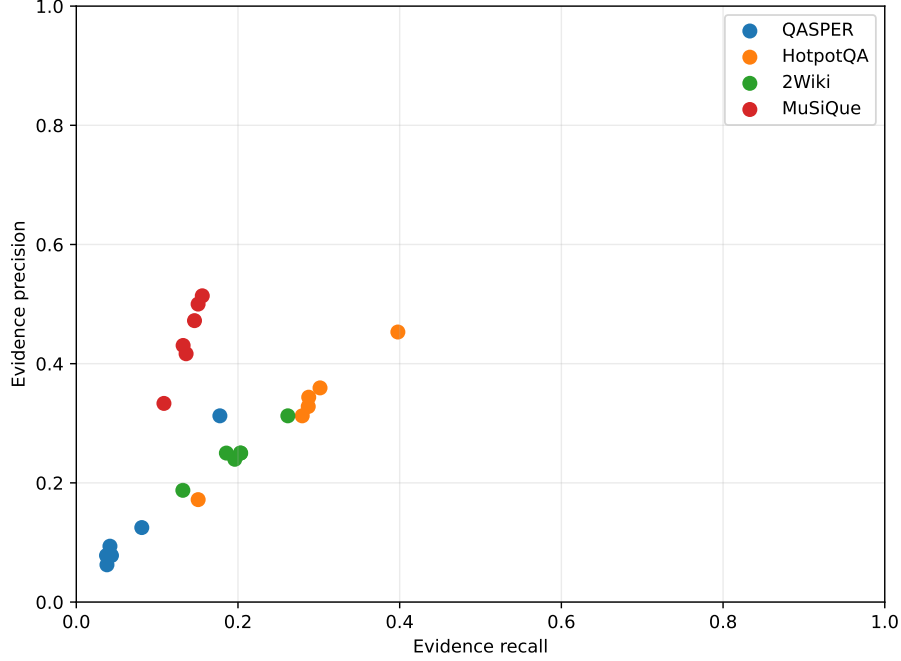


Figure 3: The channel precision–recall surface. Each point is one dataset/channel mean; higher recall obtained by fusion is not automatically a better discovery policy when precision falls.

Channel disagreement is itself diagnostic. Examples with at least the median three distinct root choices have a mean best-lexical-over-gist advantage of 0.197, compared with 0.101 below the median. High IDF-weighted query/evidence overlap similarly has a 0.200 lexical advantage, versus 0.131 in the lower half. These are post-hoc descriptive strata, not fitted dataset laws: the cohorts are too small and gold overlap is unavailable at deployment. Static hybrid remains below the best single channel in both strata (-0.124 and -0.082 recall), which argues against one uncalibrated score formula.

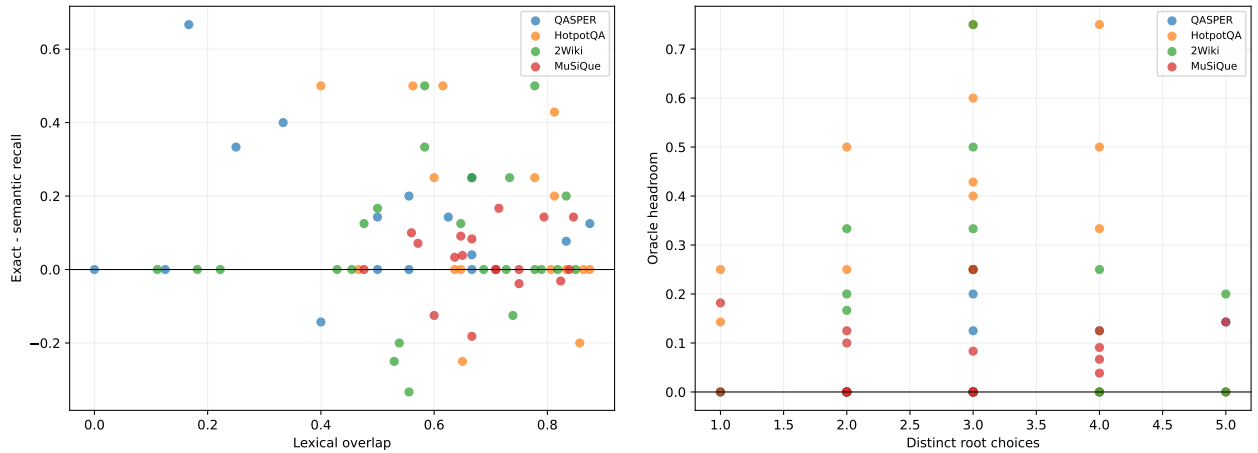


Figure 4: Left: gold query/evidence lexical overlap versus exact-over-gist recall. Right: root-channel disagreement versus validation-fixed channel-oracle headroom. Geometry is explanatory; neither variable is used as gold input to the selector.

11.2 Channel Selection and Iterative Exposure

The per-example channel oracle reaches recall 0.207 on QASPER, 0.489 on Hotpot, 0.315 on 2Wiki, and 0.199 on MuSiQue. Relative to the channel selected on validation, mean oracle headroom is 0.029, 0.338, 0.130, and 0.067. Hotpot’s large value partly reflects validation instability: gist beats BM25 by only 0.001 on eight validation examples, whereas BM25 is much stronger on test. On 2Wiki, validation selects iterative hybrid while test favors BM25. These reversals are evidence for larger calibration cohorts, not evidence that test labels should choose a production channel.

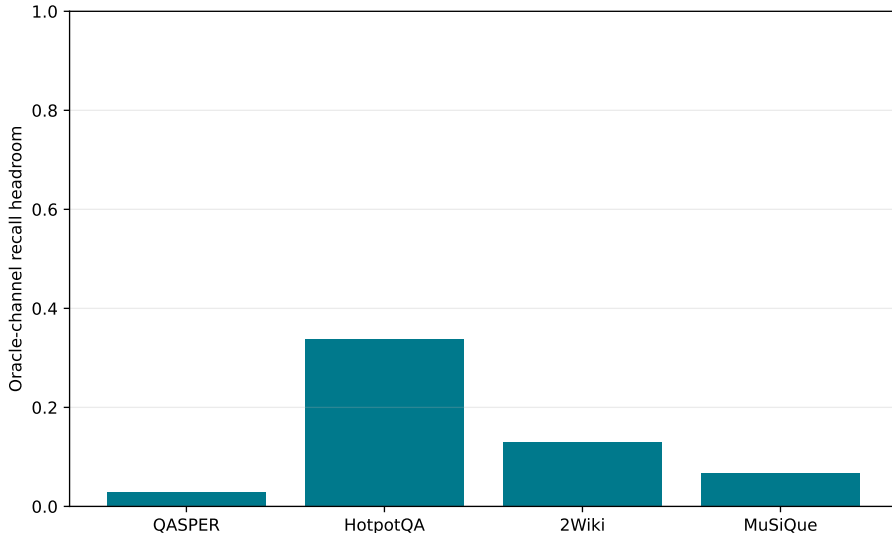


Figure 5: Per-example oracle headroom over the validation-selected fixed channel. The ceiling is useful for diagnosing adaptive opportunity, but is not a deployable result.

The fixed observable selector does not realize the oracle opportunity. Its recall is 0.081, 0.279, 0.186, and 0.146 across QASPER, Hotpot, 2Wiki, and MuSiQue. It therefore improves over static hybrid only on QASPER, ties the iterative policy elsewhere by construction, and remains below the best held-out single channel. We retain it as a leakage-audited baseline, not as the proposed final policy. A learned selector requires a larger validation cohort and should predict channel utility from query rarity, calibrated score gaps, and disagreement without using answer or evidence annotations.

A stricter new-address metric requires a normalized tokenizer piece to be absent from the question, present in an admitted hop-one chunk and unrecovered gold evidence, and rare under the example-local IDF. Such addresses occur in 6/16 QASPER, 5/16 Hotpot, 10/24 2Wiki, and 15/18 MuSiQue cases. Iterative-minus-static recall on address-positive cases is +0.100, −0.047, −0.005, and +0.003, respectively; address-negative effects are 0.000, +0.010, −0.027, and −0.042. Exposure is therefore necessary for the controlled lexical-following mechanism but not sufficient on natural data. The router must also identify the right address, preserve the right frontier, and rank the successor within the same total budget.

11.3 Separating Root and Successor Search

The preceding four-chunk result treats discovery as one decision. We next allocate two requested chunks to root search $Q \rightarrow C_1$ and two to successor search $(Q, C_1) \rightarrow C_2$. Every root channel is crossed with every successor channel for audit. The successor headline holds the root policy fixed to the validation-selected channel, so successor quality is not averaged over changing root quality.

For MuSiQue and 2Wiki, successor truth follows annotated dependency nodes; QASPER and Hotpot lack an ordered graph annotation, so successor truth is explicitly labelled as the remaining unordered evidence. This distinction prevents an inferred chain from becoming dataset truth.

Table 3: Best held-out root channel and best successor channel under the validation-selected root. Each stage requests two chunks. R/P are stage-local evidence recall/precision; comparisons are exhaustive candidate scores per example.

Dataset	Root	Root R/P	Root cmp.	Successor	Succ. R/P	Succ. cmp.	Same space?
QASPER	exact	.096/.312	143	exact new address	.025/.125	285	yes
HotpotQA	exact	.047/.094	50	native semantic	.135/.281	100	no
2Wiki	gist	.076/.125	42	exact new address	.050/.083	85	no
MuSiQue	approximate	.084/.139	82	native semantic	.032/.111	163	no

The best space changes after admission on three of four datasets. QASPER retains exact identity at both stages. Hotpot and MuSiQue prefer lexical root discovery but semantic successor expansion; 2Wiki prefers semantic entry and an exact newly exposed successor address. The low absolute recalls remain important: this is evidence that channel action should be state-dependent, not evidence that the present two-stage router solves multi-hop retrieval. Root choices also change with budget. At two root chunks, Hotpot favors exact rather than the four-chunk BM25 winner, showing that retrieval representation and conceptual budget interact.

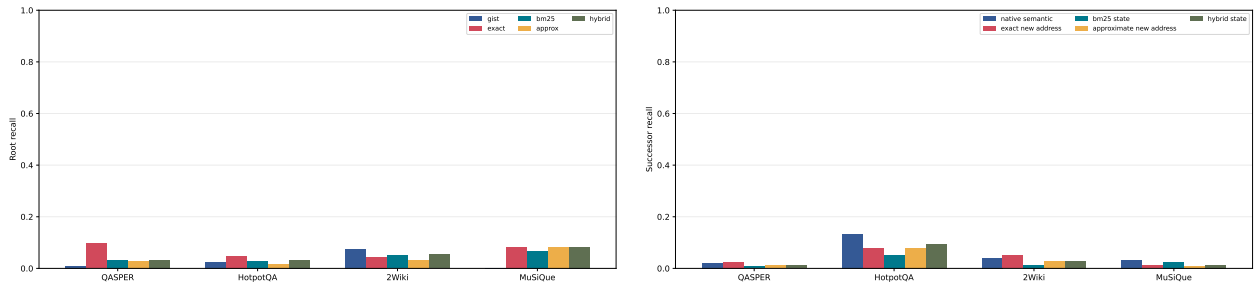


Figure 6: Root recall at two chunks (left) and successor recall at two chunks under the validation-selected root (right). Root and successor retrieval are different decisions even under one total four-chunk budget.

Per-example oracle transitions are heterogeneous. The most frequent are exact→exact (9), exact→native semantic (8), exact→BM25-state (7), gist→native semantic (6), and BM25→native semantic (5). No transition occupies a majority. The corresponding path gains are 0.038, 0.105, 0.076, 0.065, and 0.062. This is direct evidence against forcing root and successor states through one static fusion score.

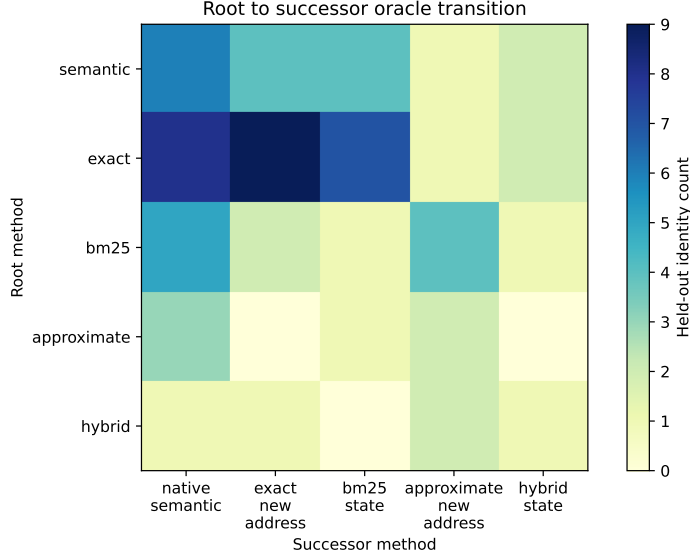


Figure 7: Final held-out per-example oracle root→successor transitions on the frozen natural cohort. Frequencies describe an upper-bound diagnostic, not a deployable controller.

11.4 True Headroom and Observable Selection

We decompose oracle gain as

$$H_{selection} = R_{oracle} - R_{best\ heldout\ fixed}, \quad H_{validation} = R_{best\ heldout\ fixed} - R_{validation\ selected}.$$

Mean $H_{selection}$ is 0.014 on QASPER, 0.021 on Hotpot, 0.053 on 2Wiki, and 0.013 on MuSiQue. Mean $H_{validation}$ is 0, 0.031, 0.026, and 0. This corrects the earlier four-chunk “oracle headroom” figure, which combined adaptive opportunity with small-validation reversal. At the two-chunk root stage, 2Wiki has the largest true adaptive opportunity, while Hotpot’s validation instability is larger than its true selector headroom.

With the validation root held fixed, successor $H_{selection}$ is 0.027, 0.028, 0.056, and 0.016; successor $H_{validation}$ is 0, 0.056, 0.011, and 0.008. Thus 2Wiki retains the largest true channel opportunity at both stages, while Hotpot’s successor-channel validation reversal is twice its true adaptive headroom. The action space is real, but these small calibration splits cannot choose it reliably.

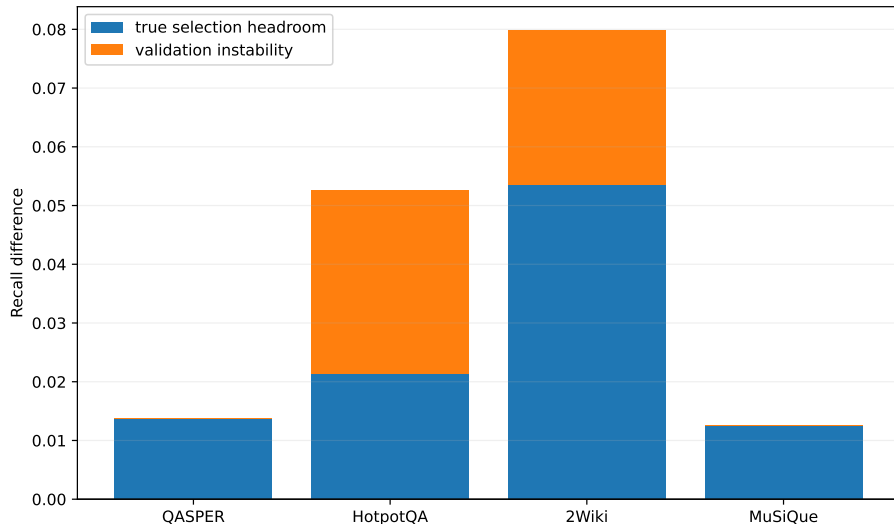


Figure 8: True per-example channel-selection headroom separated from validation-selected channel instability. Negative per-example instability is allowed; bars report means.

Neither modest selector baseline closes the gap. The fixed hand rule reaches root recall 0.020, 0.031, 0.040, and 0.073 across QASPER, Hotpot, 2Wiki, and MuSiQue. A validation-trained linear softmax selector over 24 query-only and score/rank features reaches 0.047, 0.016, 0.053, and 0.084. The best held-out fixed channels reach 0.096, 0.047, 0.076, and 0.084; the per-example oracle reaches 0.110, 0.068, 0.130, and 0.096. The linear model ties the fixed MuSiQue result and improves over the validation-selected 2Wiki channel by 0.003, but it does not justify replacing a fixed policy. No MLP is fitted because the 58 validation identities do not support a larger controller.

Root-channel disagreement also fails as a standalone selector feature. Its Pearson correlation with true per-example oracle headroom is only 0.052; selected-set Jaccard, top-10 rank overlap, and score disagreement have correlations 0.020, 0.107, and 0.013. Disagreement reveals representation diversity, but diversity alone does not say which channel is right.

11.5 Why Static Fusion Loses

At a two-chunk root budget, lexical channels add unique gold chunks relative to gist—0.812 per QASPER example, 0.188 Hotpot, 0.208 2Wiki, and 0.333 MuSiQue—but simultaneously add 3.062, 3.500, 2.792, and 2.778 unique distractors. Static hybrid recall is 0.031, 0.031, 0.053, and 0.082, below the per-example best single-channel recall of 0.110, 0.068, 0.130, and 0.087. The failure is therefore budget dilution plus score-scale mismatch, not lack of complementary evidence. Reciprocal-rank fusion avoids raw-score calibration assumptions but reaches only 0.035, 0.044, 0.053, and 0.045 recall. It is not a hidden robust winner.

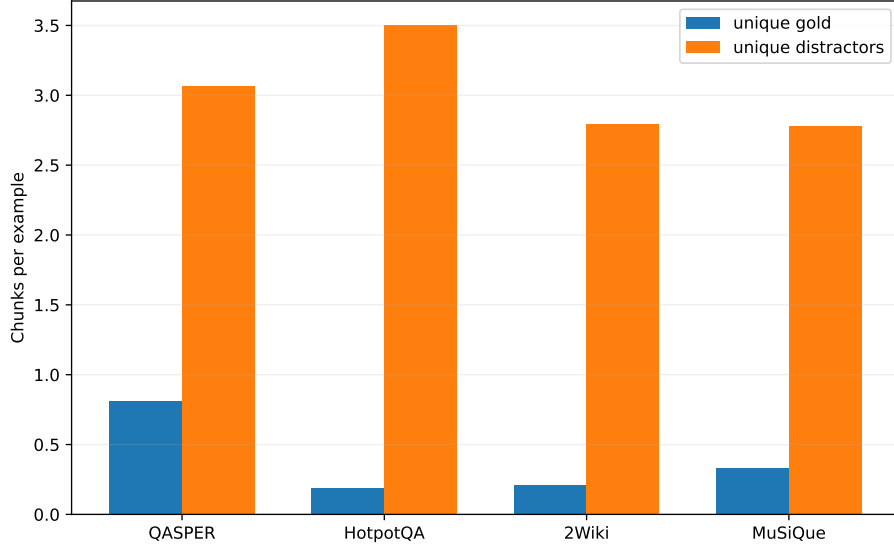


Figure 9: Unique lexical evidence and distractors relative to gist at the root stage. Complementary candidates are mostly distractors under the fixed two-chunk budget.

The inherited four-chunk MuSiQue analysis contains three cases where approximate retrieval strictly beats exact, BM25, and gist retrieval. All three have partial named-entity mentions under the observable taxonomy; this supports a partial-mention explanation on this tiny subset, not a general morphology or spelling claim. The two-chunk analysis finds no strict approximate-over-exact root win because the channels tie on the affected examples. QASPER’s three strict exact wins are classified as explicit evidence wording. The two strict Hotpot/2Wiki BM25 wins use multiple moderately informative terms rather than one dominant span. These manual-rule labels are auditable hypotheses for larger annotation, not ground truth causes.

11.6 Useful Addresses and Robustness

We tighten the iterative condition to

$$\text{UsefulAddress} = \text{Exposed} \wedge \text{GoldLinked} \wedge (\text{successor rank} \leq 4).$$

Under validation-selected roots, useful-address rates are 0.250 on QASPER, 0.063 on Hotpot, 0 on the 18 successor-bearing 2Wiki examples, and 0.222 on MuSiQue. UsefulAddress predicts a positive hybrid-state versus native-semantic path effect only on MuSiQue: +0.054 versus −0.001 without a useful address. QASPER is −0.021 in both strata; the single useful Hotpot case ties, and no 2Wiki address is competitive. Rare exposure is thus insufficient until the address also ranks within retry reach, and even the stricter predicate is dataset-dependent.

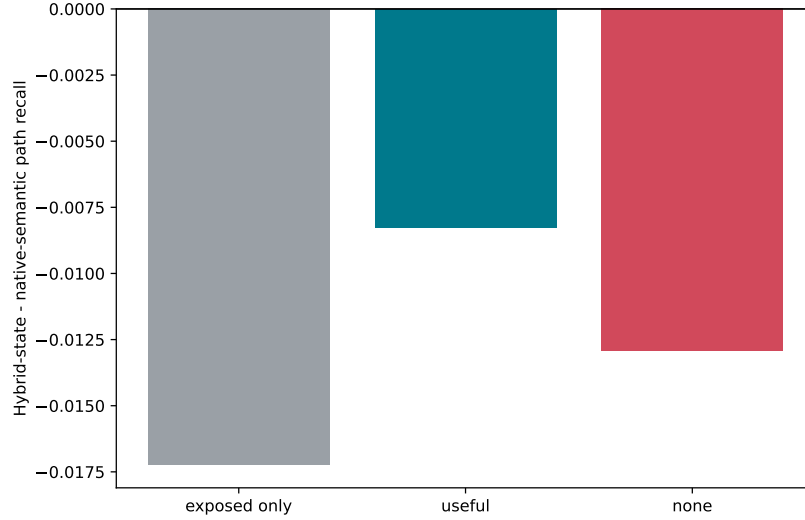


Figure 10: Path-recall gain of hybrid-state over native-semantic successor search. A competitive, gold-linked address is predictive only in the present MuSiQue subset.

The extended controlled suite separates normalization from disambiguation. All three channels recover clean, case, punctuation, near-entity, shared-prefix, numeric-ID, and URL/domain targets in all 20 trials without selecting the designated distractor. Exact matching recovers no typo targets and selects a near typo distractor in half; approximate matching recovers every typo and half of synonym/alias targets; iterative hybrid recovers half the typos but no aliases. Every channel follows the confidently wrong reference (target recovery 0, wrong-target recovery 1). Its hybrid wrong-target confidence is 0.396, below the diagnostic 0.5 action threshold, so retry is available in this constructed case, but calibration remains too weak for a deployment claim.

11.7 Controlled Chains and Failure Controls

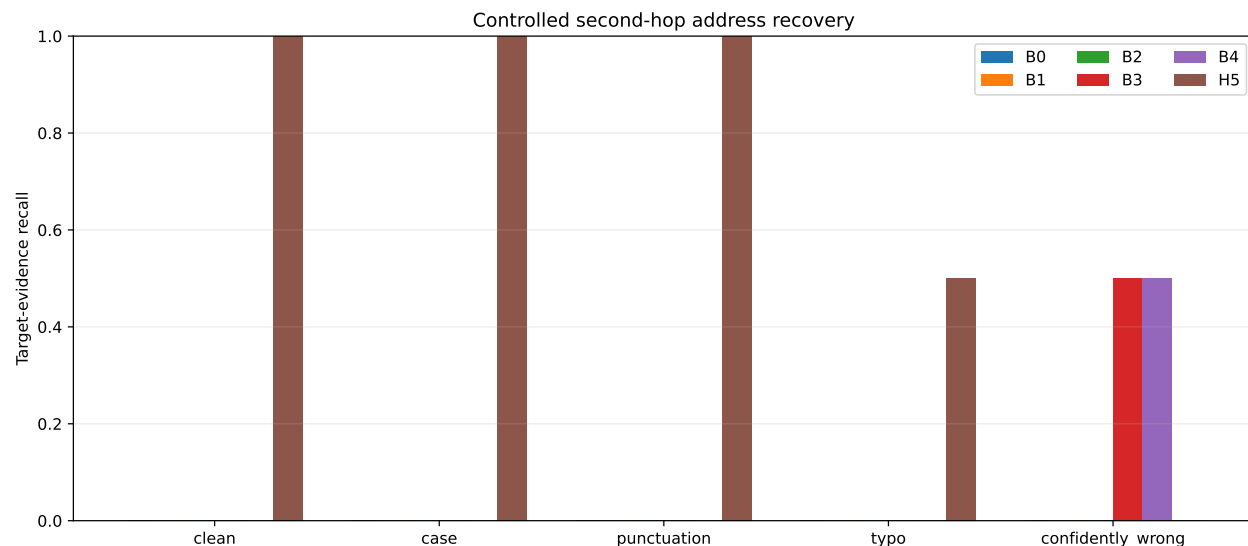


Figure 11: Controlled second-hop target recovery at a two-chunk request budget. H5 recovers every clean, case, and punctuation target, half of typo targets, and none when the exposed lexical address is confidently wrong.

Gist-only retrieval recovers none of the hidden targets because the target sits behind stronger semantic distractors. H5 recovers 100% of clean, case, and punctuation targets, showing that admitted evidence can expose a useful later-hop address. Recovery falls to 50% for one-character perturbations under the current approximate score. In the contradictory condition, H5 follows the wrong high-overlap address and recovers no targets; wrong selected candidates receive mean raw confidence 0.585. This is catastrophic rather than graceful degradation and prevents confidence-gated deployment without stronger disambiguation.

11.8 Confidence and Cost

Validation-fitted calibration remains weak. On held-out candidates, H5 has Brier score 0.185 and ECE 0.151, but no candidate crosses the 0.5 action threshold. BM25 has Brier 0.205 and ECE 0.096, and likewise has zero coverage at 0.5. Exact matching is particularly overconfident (Brier 0.281, ECE 0.251). Provenance is therefore useful diagnostically, but the present calibrator is not a production materialization controller.

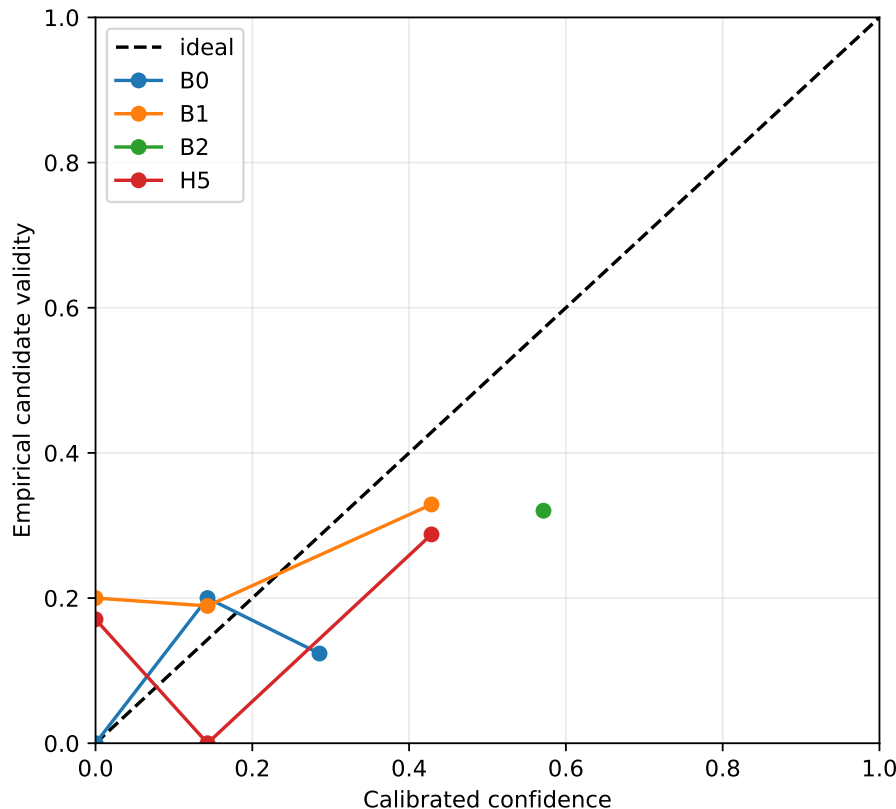


Figure 12: Held-out reliability after validation-only histogram calibration. Sparse and collapsed confidence levels limit threshold usefulness; these curves are diagnostics, not evidence of deployable calibration.

The original discovery run logs logical, resident, requested, and materialized counts separately. B0–H5 request exactly four chunks per natural example; materialized chunks and tokens are zero by construction in that run. Averaged over the held-out examples, B0 takes 5.56 ms to reconstruct the gist index and 1.63 ms to route (7.19 ms cold total). B1 adds 34.42 ms of token-index construction and takes 62.60 ms to route (102.42 ms cold). Two-hop H5 adds 35.09 ms of token-index construction and takes 181.59 ms to route (224.46 ms cold), performing 289 mean semantic and 289 token comparisons versus B0’s 96 semantic comparisons. These are exhaustive Python research timings retained for historical comparison; the indexed confirmation below supersedes them as the systems path.

12 Final Stability and Referential-Validity Diagnostics

12.1 Cohort Feasibility and Selection Stability

The frozen study contains 132 identities: 58 validation and 74 held out. The held-out composition is 16 QASPER, 16 HotpotQA, 24 2Wiki, and 18 MuSiQue identities. We sought a cheap expansion to at least 50 identities per dataset, but no larger identity-compatible feature capture exists for the fixed model revision, layer, chunking, evidence map, and query aggregation. New hidden-state extraction would change the provenance of this final iteration. We therefore retain all 132 identities and report 1,000 deterministic, identity-paired bootstrap resamples at increasing cohort

sizes. Resamples estimate uncertainty; they are not new examples.

At each dataset’s full held-out size, QASPER chooses exact root search in 94.3% of resamples. Hotpot chooses exact in 60.6%, with BM25 at 23.1% and semantic at 15.3%. 2Wiki chooses semantic in 66.6%. MuSiQue remains unresolved: approximate wins 36.8%, hybrid 35.4%, and BM25 24.8%. Successor selection is most stable for Hotpot native semantic (93.2%). QASPER splits between exact-new-address (57.7%) and native semantic (38.3%); 2Wiki splits among exact-new-address (47.8%), hybrid-state (26.4%), and native semantic (25.2%); MuSiQue splits between native semantic (59.0%) and BM25-state (34.3%). Thus the three-of-four root/successor mismatch survives resampling, but several channel labels should be read as cohort preferences rather than settled rankings.

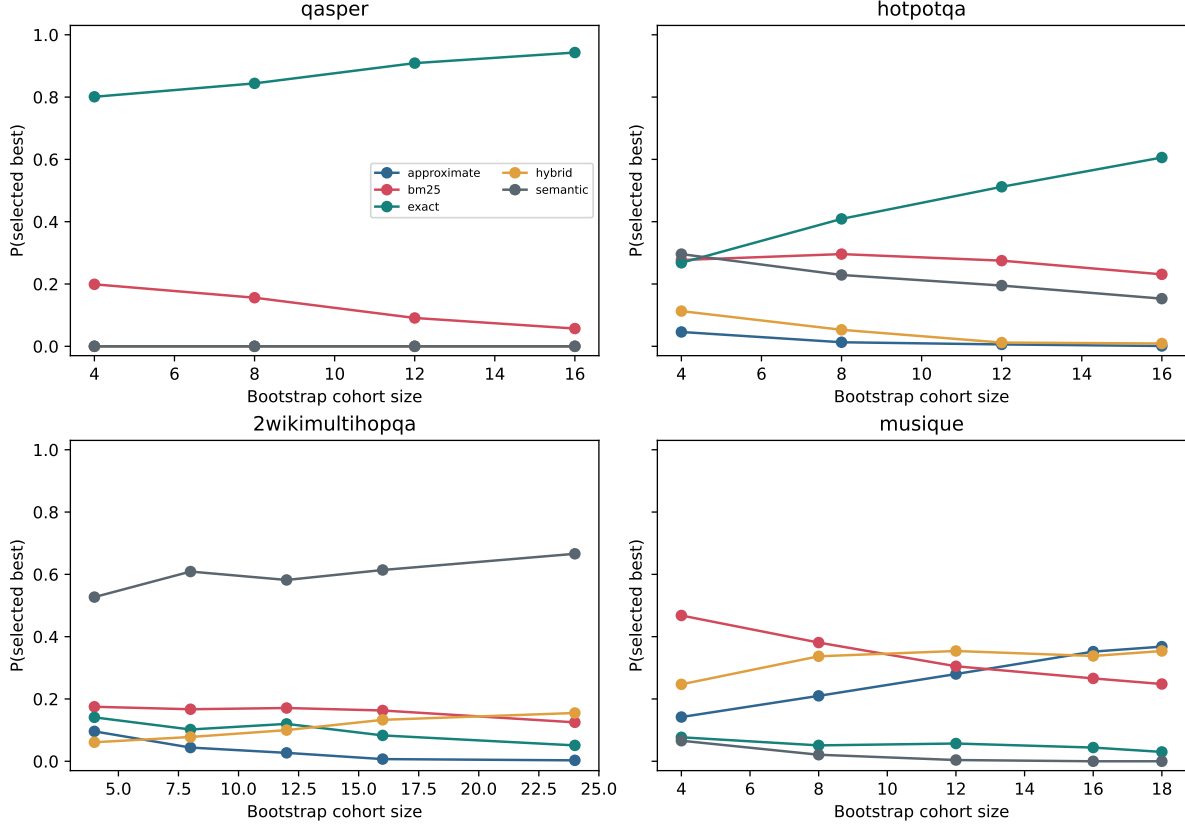


Figure 13: Probability that each root channel is selected as best under identity-paired bootstrap resampling. The endpoint uses every held-out identity available for that dataset.

The headroom decomposition remains unchanged in meaning. Root $H_{selection}$ ranges from .013 to .053 and successor $H_{selection}$ from .016 to .056. The paired 95% interval excludes zero only for 2Wiki root, [.014, .099], and MuSiQue successor, [.002, .035]. This is evidence for heterogeneous action opportunity, not evidence that a deployable selector has been learned. $H_{validation}$ remains separately visible and is especially large for Hotpot successor search.

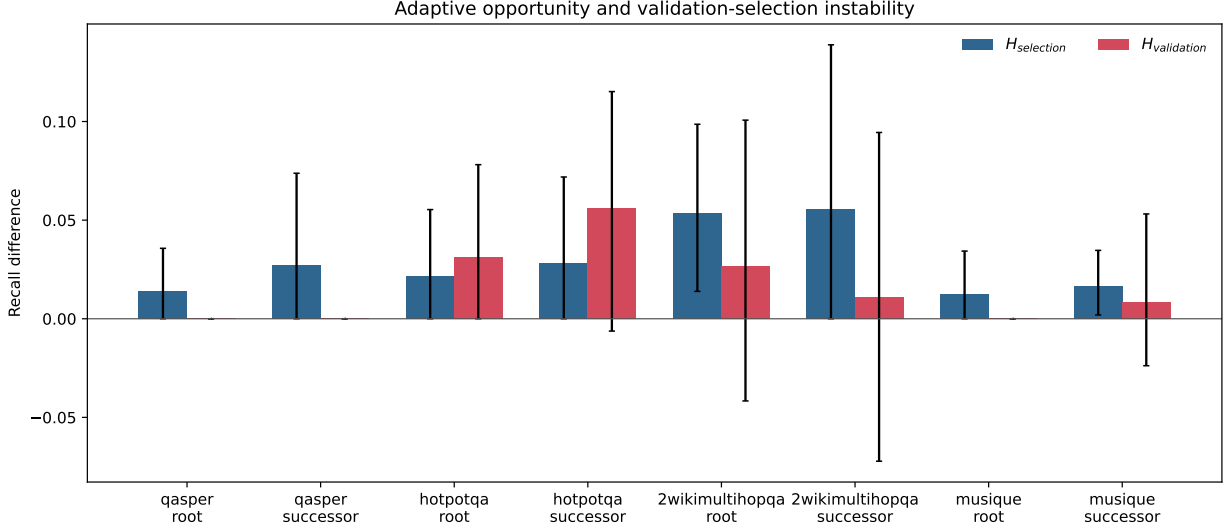


Figure 14: Per-example adaptive opportunity $H_{selection}$ and validation-selection instability $H_{validation}$ with paired-bootstrap 95% intervals. Root and successor components are estimated separately.

12.2 Match Confidence Is Not Referential Validity

The replay exports 1,255 candidate-level confidence rows without changing any route. Signals include top score, top-1/top-2 gap, score entropy and effective support, exact span length, matched-token rarity and occurrence count, approximate similarity, ambiguity count, alias indication, cross-channel agreement, semantic consistency, current-path consistency, and source consistency. Gold membership is added only as the evaluation outcome. Dataset identity, answer text, evidence geometry, and oracle labels are forbidden selector inputs.

Validation-CDF calibration does not turn these signals into reliable referential probabilities. For the composite referential-consistency proxy, root-channel AUROC ranges from .353 to .605 and averages .424; successor-channel AUROC ranges from .396 to .717 and averages .526. Mean AUPRC is .105 at root and .156 at successor, while mean ECE is .401 and .456. The apparently stronger BM25-state AUROC (.717) is based on only four positive top-1 successors among 68 held-out identities and does not support a general calibration claim. Score strength and referential validity remain different quantities.

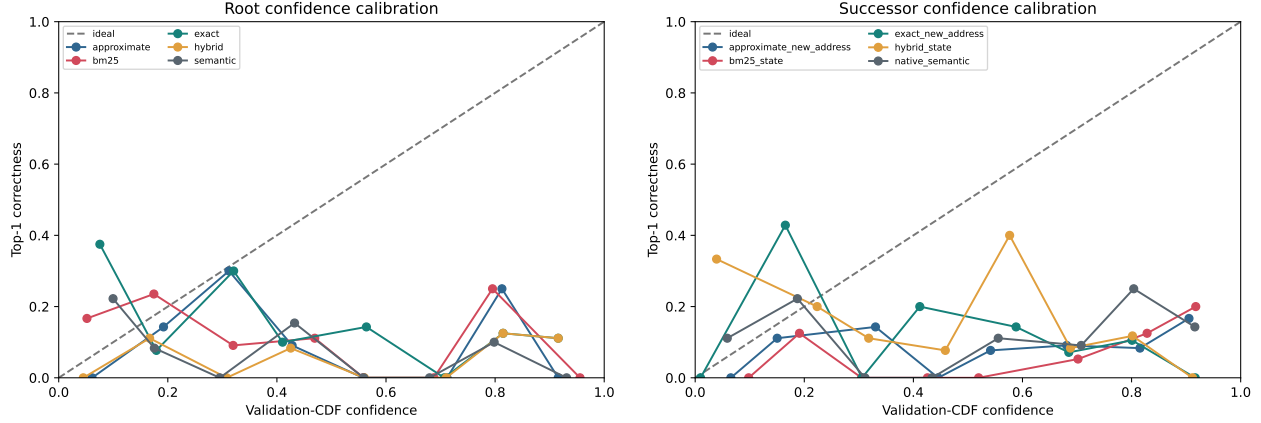


Figure 15: Held-out top-1 correctness against validation-CDF referential-confidence bins. Sparse positives and non-monotone curves prevent a deployment interpretation.

A diagnostic consistency gate trusts the highest-confidence lexical candidate only when its semantic compatibility exceeds a validation-selected threshold; otherwise it falls back to semantic search. It never improves root top-1 correctness over lexical confidence alone. At successor search it is also inconsistent: on Hotpot it falls from .312 for semantic alone to .062, and on 2Wiki from .111 for lexical confidence alone to .056. Static fusion remains non-dominant. The gate is useful as a falsified simple hypothesis, not a new hybrid algorithm.

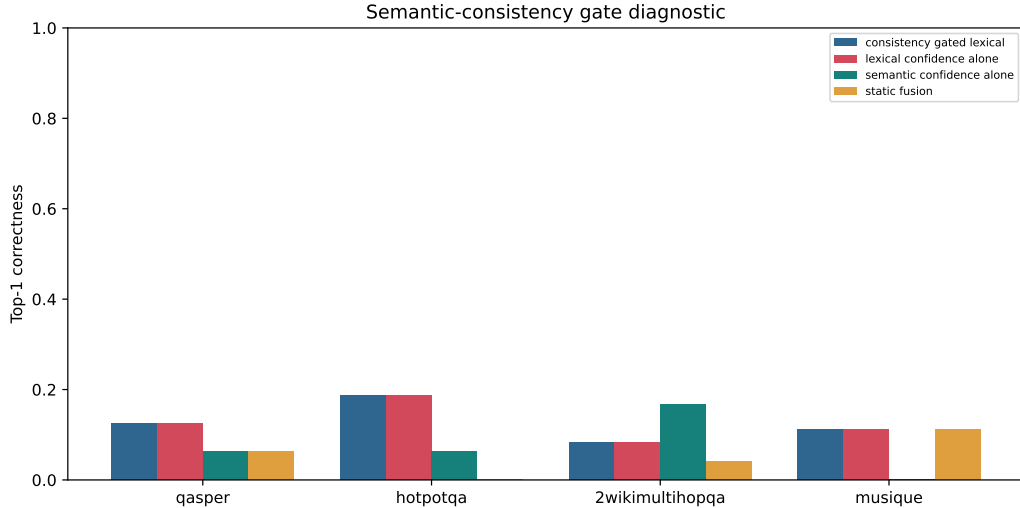


Figure 16: Root top-1 correctness for lexical confidence, semantic confidence, a validation-thresholded consistency gate, and static fusion. The gate does not improve the lexical decision on any cohort.

12.3 Ambiguity, Abstention, and Useful Addresses

The expanded controlled suite preserves the clean, case, punctuation, typo, alias, near-entity, shared-prefix, numeric-ID, URL, and confidently-wrong conditions and adds five referential conflicts. Stale aliases route to the wrong target in every trial for exact, approximate, and hybrid search. A same-class wrong instance is selected in every approximate trial and half of exact and hybrid trials. Under the correct-entity/wrong-relation fixture, approximate search splits evenly

between valid and wrong targets; exact and hybrid recover neither. Two plausible references happen to select the valid candidate because of deterministic tie order, which demonstrates neither disambiguation nor robustness. Approximate matching still repairs every typo and half of the alias cases, but its gain comes with collision risk.

Confidence-based rejection creates an opportunity to avoid some controlled wrong routes, particularly for approximate search, but the curve is fixture-derived and thresholds are not deployment-calibrated. We therefore export the retained/rejected decisions to Paper 3.5 rather than claiming a safe abstention policy.

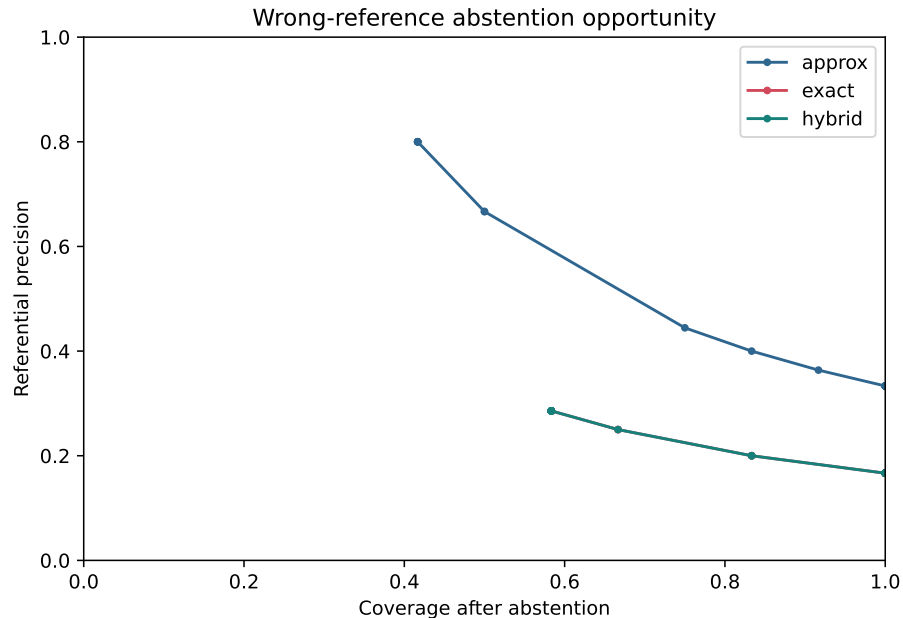


Figure 17: Controlled referential precision versus retained coverage as wrong-target confidence is thresholded. This is an abstention opportunity bound, not a natural-cohort policy result.

The observable UsefulAddress proxy combines address presence, rarity, uniqueness, exact/approximate confidence, and semantic consistency. On 68 held-out successor-bearing identities it attains AUROC .555 and AUPRC .230 for the nine true UsefulAddress cases. Its validation-selected threshold retains 70.6% of examples at precision .125 and recall .667, rejecting 17 wrong and three correct cases. This is weak screening signal, not a substitute for the gold diagnostic or a retry controller.

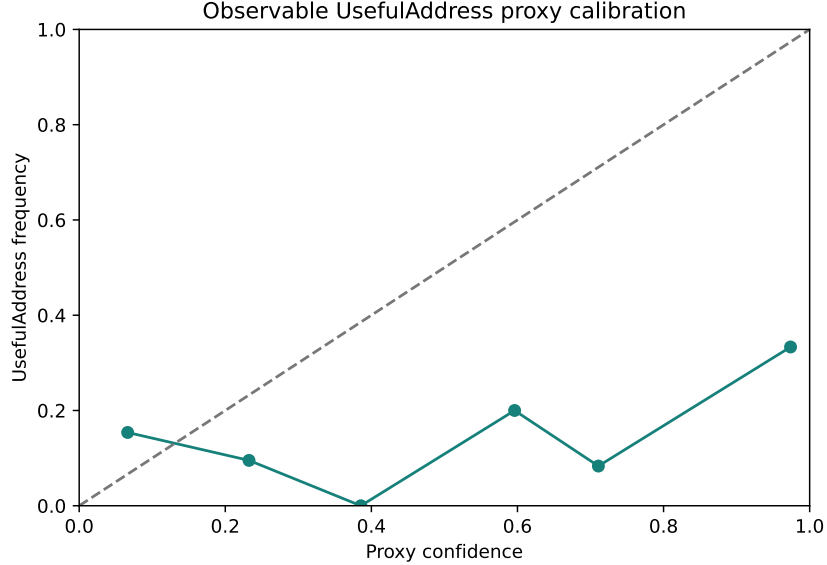


Figure 18: Calibration of the deployment-observable UsefulAddress proxy. The proxy is only weakly informative and remains unsuitable as a standalone action gate.

13 Indexed and End-to-End Confirmation

13.1 Production-Index Scaling

We generated aligned sidecars containing 256, 1,024, and 4,096 chunks for both the Qwen3-0.6B and SmolLM2-135M tokenizers. Each query was measured exhaustively and through a 64-row indexed pool on an Intel i7-6700HQ CPU with Python 3.10.11 and PyTorch 2.12.1. The indexed path reproduced the exhaustive top four in every paired query. Table 4 reports warm single-query measurements at 4,096 chunks. The candidate fraction is $64/4,096=1.56\%$; it is a work bound, not the fraction ultimately materialized.

Table 4: Warm routing at 4,096 chunks. Times are milliseconds per query; speedup compares the same Python channel implementation before and after posting-list narrowing. Index MiB is serialized payload size and excludes Python-object overhead.

Tokenizer	Channel	Exhaustive	Indexed	Speedup	Index MiB
Qwen	exact	33557	119	283×	10.06
Qwen	weighted	31375	530	59×	10.06
Qwen	approximate	31141	499	62×	10.06
Qwen	cascade	30965	556	56×	10.06
SmolLM	exact	37048	125	300×	10.94
SmolLM	weighted	37440	649	58×	10.94
SmolLM	approximate	37005	632	59×	10.94
SmolLM	cascade	36886	577	64×	10.94

At 4,096 chunks, cold construction cost 3.37 s for Qwen and 2.76 s for SmolLM. Adding 64 chunks by immutable full-statistics rebuild took 0.80 s for either tokenizer. These conservative research implementations are a substantial improvement over exhaustive Python scoring, but the

0.50–0.65 s fuzzy paths are not yet serving-ready. The measurements support candidate-narrowing and parity claims, not competitive production latency.

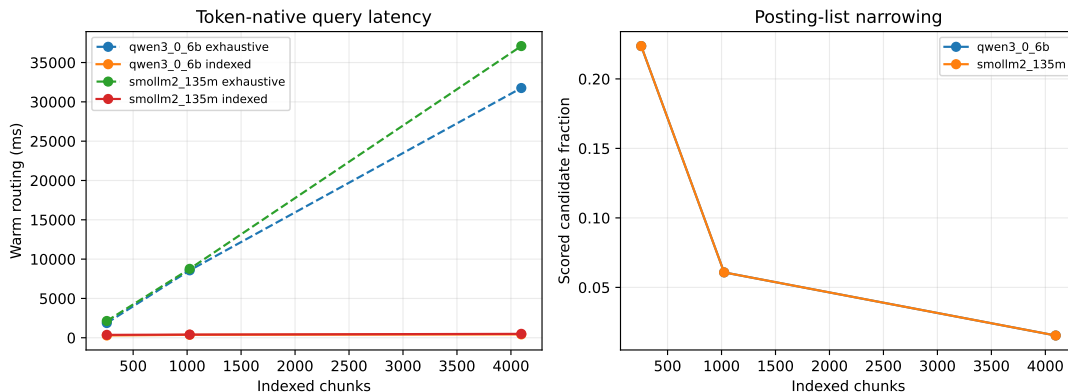


Figure 19: Exhaustive and indexed routing latency. The indexed candidate pool stays bounded while the logical corpus grows.

13.2 Tokenizer and Surface-Form Replication

Across 13 controlled perturbations, cascade recall@4 was 0.985 with corpus IDF and 0.969 with either no stop suppression or the fixed list, identically for both tokenizers. Exact matching reached 0.923, token n -grams 0.754, approximate matching 0.262, and bounded edit matching 0.246. Thus the cascade is portable across these two tokenizers because its stages compensate for different failure modes; the weak fuzzy channels do not by themselves solve referential ambiguity.

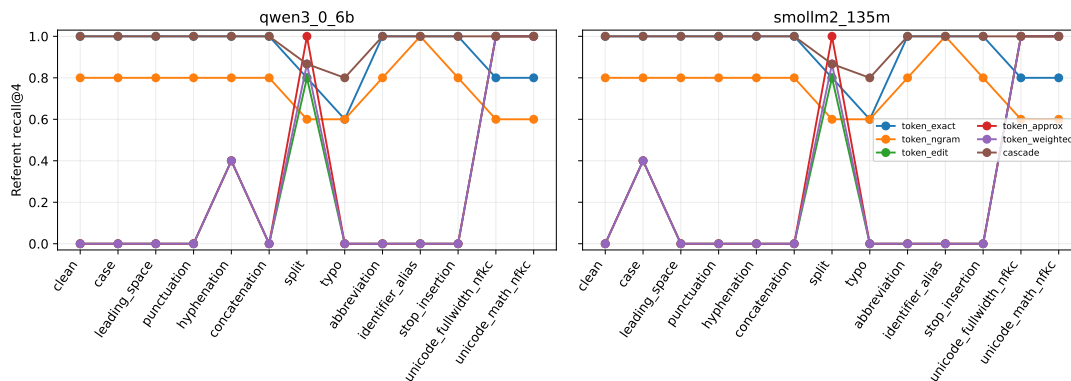


Figure 20: Controlled target recovery under tokenizer and surface-form perturbations. NFKC normalization covers compatibility forms; exact identity still fails under genuinely changed token sequences, where the cascade supplies fallback.

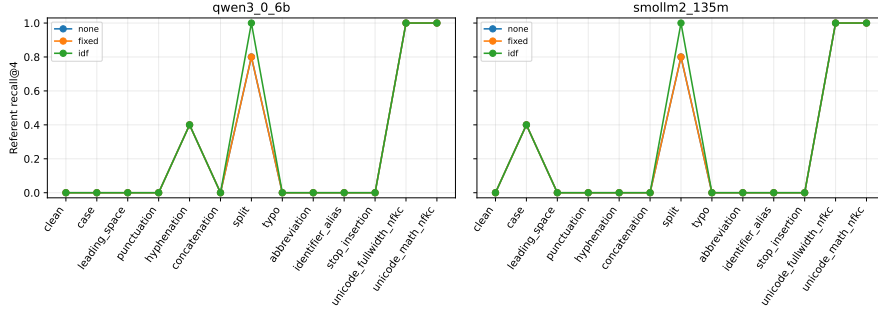


Figure 21: Fixed suppression versus corpus-IDF weighting. The small IDF advantage supports a corpus-aware default but does not materially change the weak standalone weighted channel.

13.3 Native-K/V Gate B Protocol

The confirmation uses two frozen model/tokenizer families and identities that are disjoint from the original 24-example-per-dataset Paper 2.6 cohort. For each model, 20 new examples per dataset tune only the observable channel selector, and 50 further examples per dataset form the held-out test. A serialized audit confirms zero pairwise overlap among the 48 legacy, 40 validation, 100 test, and 16 interaction identities. For this mechanism check, annotated evidence strings are placed first, followed by the original source, and the result is bounded to 1,536 tokens. Gold labels are not passed to any learned or fixed channel, but this construction is annotation dependent and is not an unmodified QA benchmark. The bounded source is partitioned into 32-token chunks, and all routed conditions receive four chunks and at most 128 physical native-K/V tokens across four late consumption layers. Semantic, exact, BM25, approximate, hybrid, a five-seed validation-trained selector, and a padded evidence oracle share this budget.

Generation controls hold the prompt and frozen model fixed: disabled and empty use no memory; shuffled uses the selected budget from another location; irrelevant uses non-evidence chunks; and bounded direct context places the same bounded source in the ordinary prompt. We report evidence recall/precision, EM, token F1, mean gold-answer token log probability, generation time, physical K/V counts, and paired bootstrap intervals. The oracle is an annotated diagnostic ceiling and never participates in channel training.

13.4 Gate B Results

Tables 5 and 6 report the complete matched-budget channel comparison. The cross-family retrieval result is unusually consistent: BM25 has the highest Hotpot recall (0.665 Qwen; 0.662 SmolLM), while approximate token matching has the highest QASPER recall (0.747; 0.783). The validation-trained adaptive selector does not beat those fixed channels. Across its five initialization seeds, recall ranges only 0.540–0.560 on Qwen/Hotpot and 0.556–0.576 on Qwen/QASPER, so its deficit is not an unlucky seed.

Table 5: Qwen3-0.6B Gate B results, 50 held-out examples per dataset. $\Delta \log p$ is mean gold-answer token log probability relative to disabled PRA. Every listed condition uses at most four chunks/128 native-K/V tokens.

Condition	HotpotQA			QASPER		
	Recall	$\Delta \log p$	F1	Recall	$\Delta \log p$	F1
semantic	.473	.744	.108	.364	-.298	.086
exact	.482	.534	.074	.619	-.148	.095
BM25	.665	.545	.113	.228	-.194	.100
approximate	.593	.656	.101	.747	-.013	.094
hybrid	.505	.518	.120	.457	-.231	.098
adaptive	.543	.528	.094	.576	-.081	.082
oracle	1.000	1.088	.103	1.000	.090	.098

Table 6: SmolLM2-135M Gate B results under the same identities, budgets, and frozen protocol.

Condition	HotpotQA			QASPER		
	Recall	$\Delta \log p$	F1	Recall	$\Delta \log p$	F1
semantic	.320	.185	.033	.348	.056	.024
exact	.450	.127	.036	.609	-.091	.020
BM25	.662	.152	.042	.328	-.062	.014
approximate	.593	.166	.032	.783	-.008	.018
hybrid	.422	.115	.034	.437	-.018	.028
adaptive	.518	.151	.028	.446	-.055	.016
oracle	1.000	.313	.047	1.000	-.008	.060

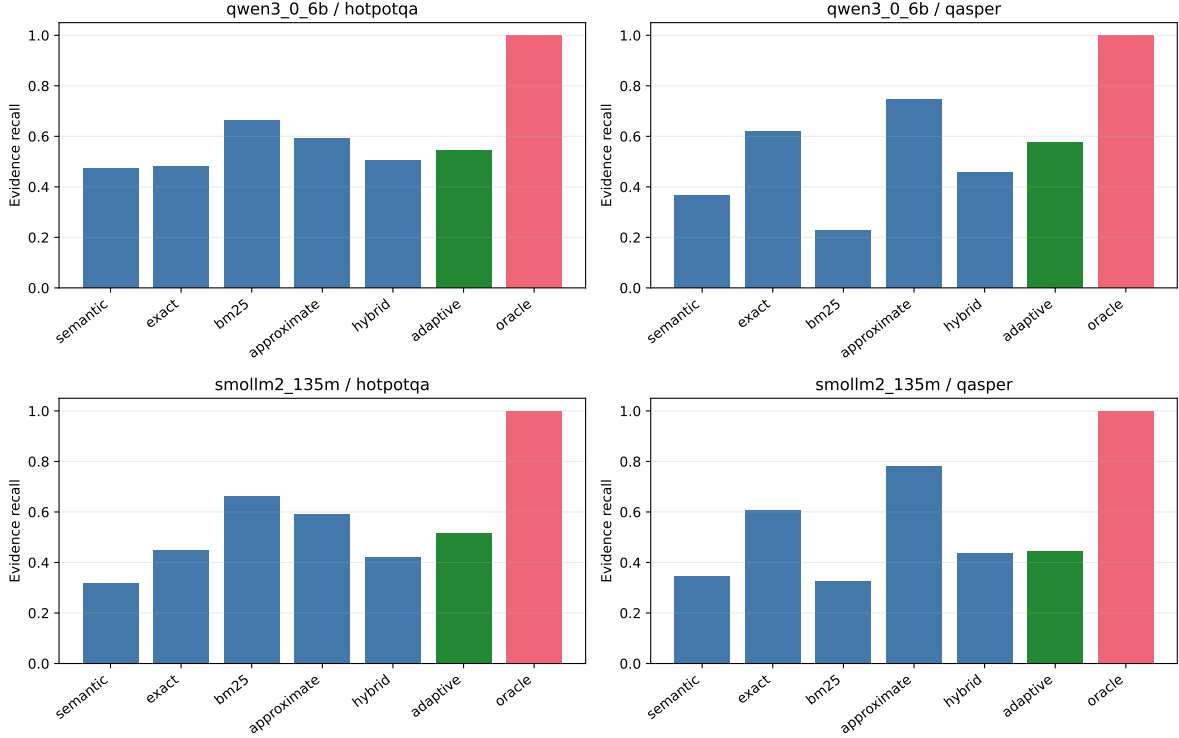


Figure 22: Cross-model held-out evidence recall. Dataset-level channel geometry replicates, while the observable selector remains below the best fixed channel.

Hotpot gives the resolved consumption result. Relative to disabled PRA, Qwen semantic routing improves gold-answer log probability by 0.744 (paired bootstrap 95% CI [0.414, 1.132]); SmolLM improves by 0.185 [0.038, 0.345]. More stringently, the matched oracle beats irrelevant memory by 0.770 [0.508, 1.052] for Qwen and 0.256 [0.163, 0.354] for SmolLM. Correct evidence therefore matters after materialization rather than merely changing logits because additional K/V exists.

QASPER separates discovery from use. Approximate routing recovers roughly three quarters of evidence for either model, yet its $\Delta \log p$ versus disabled is unresolved: -0.013 [-0.190, 0.185] for Qwen and -0.008 [-0.138, 0.123] for SmolLM. The oracle still beats irrelevant memory by 0.154 [0.074, 0.231] and 0.091 [0.033, 0.150], respectively. Thus the native-K/V mechanism discriminates correct from wrong memory, but no fixed QASPER channel establishes a benefit over no memory. This is a routing-to-consumption gap, not evidence that retrieval recall is false.

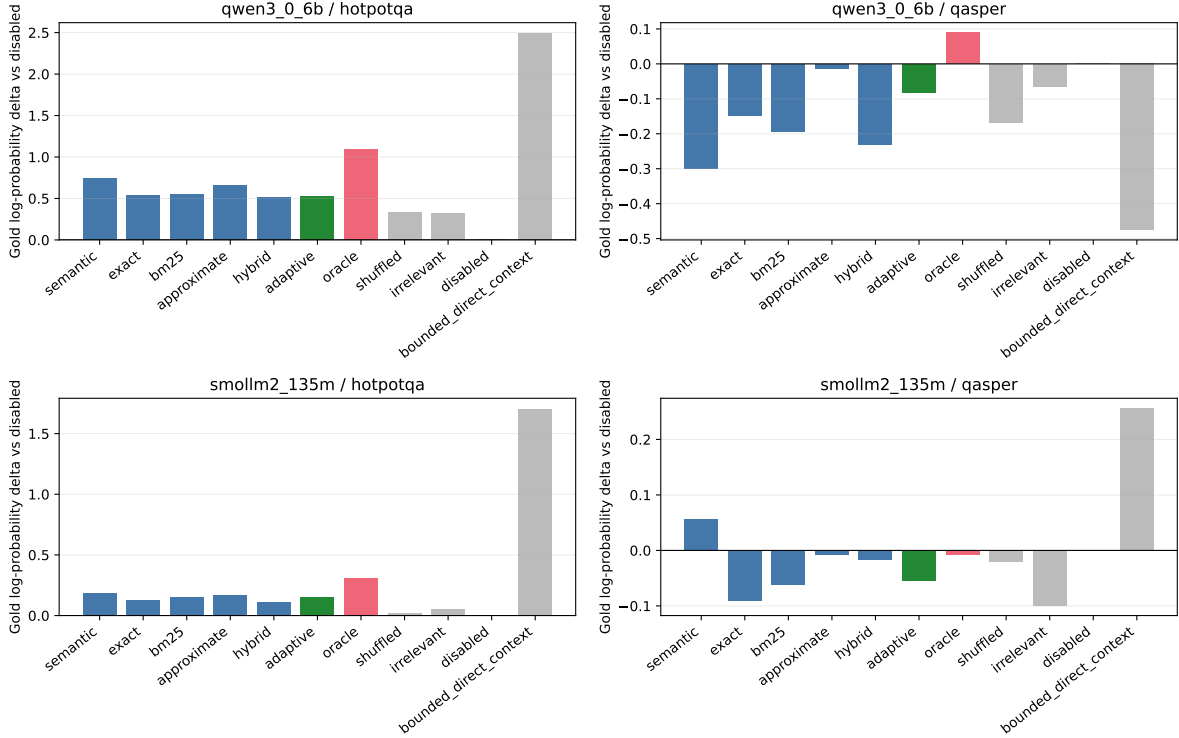


Figure 23: Gold-answer likelihood under matched discovery, oracle, wrong-memory, disabled, and bounded direct-context conditions. Direct context is a compute-heavier comparison, not a matched K/V condition.

Physical accounting confirms the match. Routed conditions materialize 127.8–128.0 unique tokens, 8.33–9.36% of the bounded source, at each of four late layers. This is about 512 layer-token states: 1.05 MiB of fp16 K/V for Qwen and 0.39 MiB for SmolLM under their respective grouped-query head geometry. No disabled or empty row reports active K/V.

Generated-answer EM/F1 does not provide a resolved positive result. Qwen Hotpot memory conditions reach EM 0.00–0.02 and F1 0.074–0.120; SmolLM EM is zero. Bounded direct context reaches Qwen Hotpot F1 0.183, but costs 5.28 s generation per example versus about 1.4 s for 128-token PRA memory. The corresponding SmolLM times are 2.63 s versus about 1.3 s. These tiny frozen models make teacher-forced likelihood the more sensitive causal measure; the paper does not relabel it as answer-quality improvement.

13.5 Channel, Granularity, Budget, and Layer Interactions

We next cross eight channels with 16-, 32-, and 64-token routing chunks; Top-2, Top-4, and Top-8 selection; and an early versus late routing layer. The fresh exploratory cohort contains eight HotpotQA and eight QASPER identities, disjoint from every legacy, validation, and Gate B test identity, for each model. The implementation makes each requested budget reachable by setting per-hop branch and beam width to 1, 2, and 4, respectively. The resulting 4,608 rows select exactly 2, 4, or 8 chunks in the corresponding condition. This explicit reachability check corrects a pilot in which Top-8 was silently capped by a narrower beam.

At 32-token chunks, increasing the physical budget gives a consistent coverage–precision exchange. Averaged over channels, late-layer Hotpot recall rises from 0.562 to 0.703 to 0.883 for

Qwen and from 0.547 to 0.703 to 0.953 for SmolLM as the budget moves from two to four to eight chunks. Complete-evidence recovery rises from 0.406 to 0.562 to 0.812 and from 0.406 to 0.547 to 0.906, while precision falls from 0.414 to 0.229 and from 0.375 to 0.225. QASPER follows the same broad pattern: recall reaches 0.917 for Qwen and 0.943 for SmolLM at Top-8, but precision declines relative to Top-4. Breadth is therefore useful for evidence discovery, but it is not free selectivity.

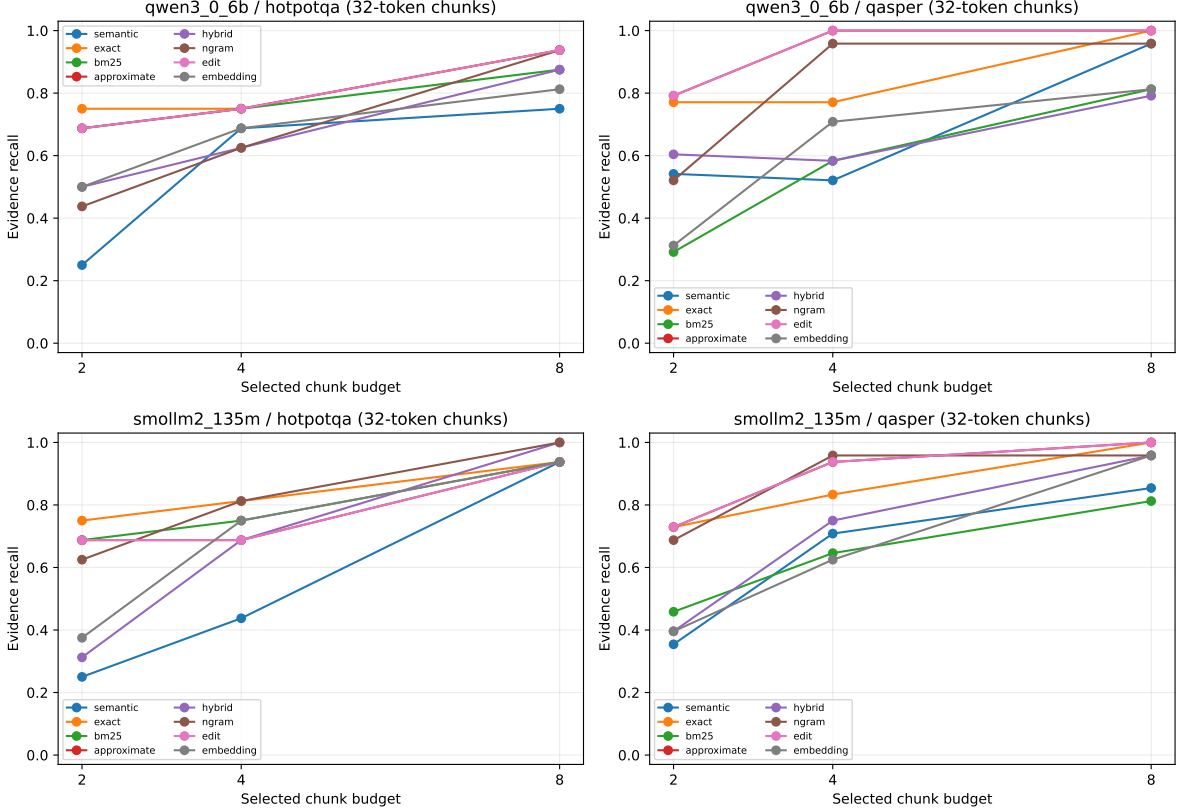


Figure 24: Channel-specific evidence recall as the reachable selected-chunk budget increases at fixed 32-token granularity. Larger budgets improve coverage but also materialize more distractors.

Holding routed physical tokens fixed at 128 reveals a distinct granularity effect (Table 7). The best channel can change with chunk size and model: for example, Qwen/Hotpot favors token n -grams at 16 and 64 tokens but bounded edit matching at 32; SmolLM/QASPER favors hybrid at 16, token n -grams at 32, and exact matching at 64. These are exploratory maxima over eight examples, not a new universal policy. Their value is diagnostic: channel choice cannot safely be tuned independently of the chunker.

Table 7: Best late-layer channel at a matched 128 routed tokens. Chunk count changes inversely with chunk size. C_E is complete-evidence recovery.

Model	Dataset	Chunk	Best channel	Recall	Precision	C_E
Qwen	HotpotQA	16	token n -gram	.938	.266	.875
Qwen	HotpotQA	32	bounded edit	.750	.281	.625
Qwen	HotpotQA	64	token n -gram	.875	.438	.875
Qwen	QASPER	16	token n -gram	1.000	.547	1.000
Qwen	QASPER	32	bounded edit	1.000	.812	1.000
Qwen	QASPER	64	exact	1.000	.938	1.000
SmolLM	HotpotQA	16	bounded edit	.938	.328	.875
SmolLM	HotpotQA	32	exact	.812	.312	.625
SmolLM	HotpotQA	64	token n -gram	1.000	.500	1.000
SmolLM	QASPER	16	hybrid	.958	.438	.875
SmolLM	QASPER	32	token n -gram	.958	.625	.875
SmolLM	QASPER	64	exact	1.000	.875	1.000

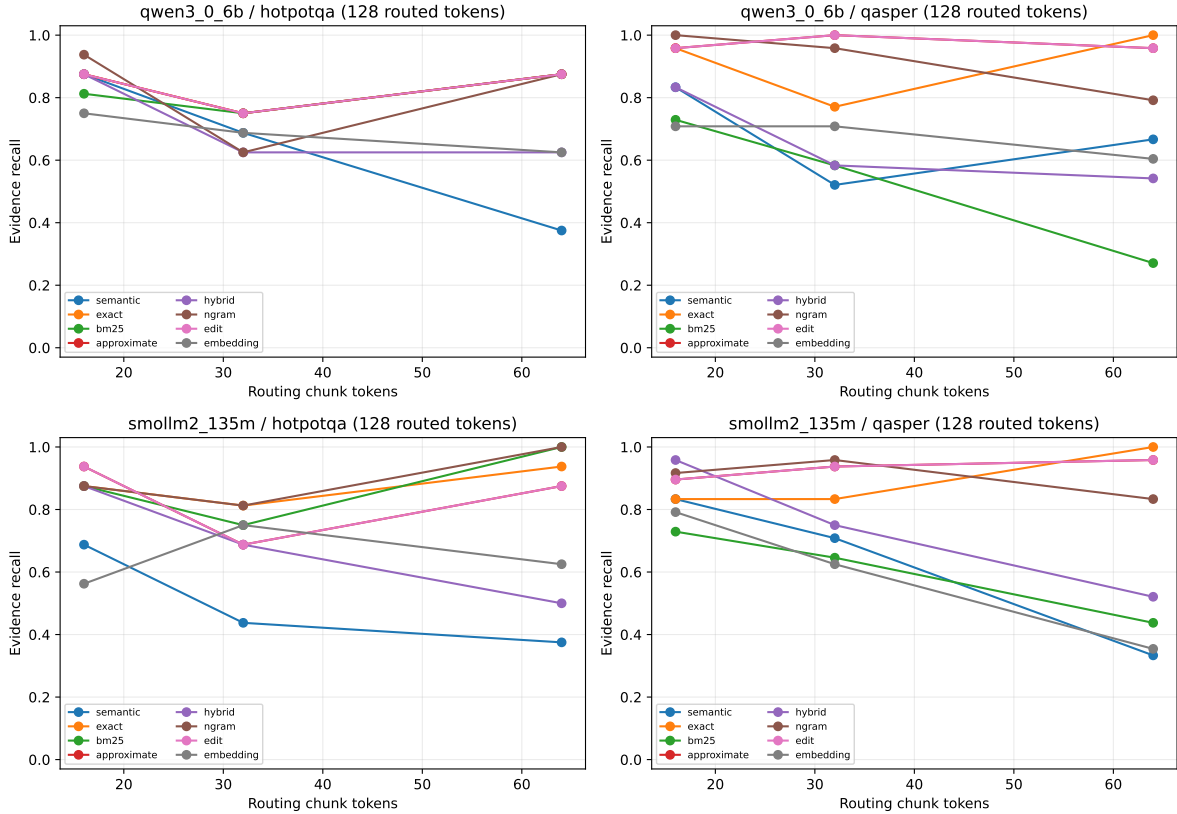


Figure 25: Evidence recall across channel and chunk granularity at a fixed 128 routed physical tokens. The ordering changes across datasets, models, and chunk sizes.

Layer sensitivity is channel specific. Token-native exact, BM25, approximate, n -gram, edit, and embedding scores are layer invariant by construction. Moving semantic routing from the early to late measured layer changes mean recall by +0.174 for Qwen/Hotpot, +0.046 for Qwen/QASPER, +0.021 for SmolLM/Hotpot, and -0.039 for SmolLM/QASPER; hybrid changes by +0.090, -0.021, -

0.028, and -0.009. Late hidden states are therefore not automatically better routing representations. The small interaction cohort is evidence for coupling among channel, granularity, budget, and layer, not a replacement for the 50-example Gate B comparison.

13.6 Natural Wrong-Reference Calibration

Naturally annotated evidence spans label whether each channel’s top chunk is relevant. Approximate top-one accuracy on Hotpot is 0.94 for both models, but falls to 0.36–0.40 on QASPER. Exact reaches 0.64 on Hotpot and 0.46–0.54 on QASPER; BM25 reaches 0.48–0.54 and 0.08. Validation-CDF thresholds chosen for at least 0.80 precision do not transfer: except for approximate Hotpot (0.94 precision), held-out precision remains below 0.64 and often below 0.20. Confidence continues to be a referential-validity problem even when recall is strong.

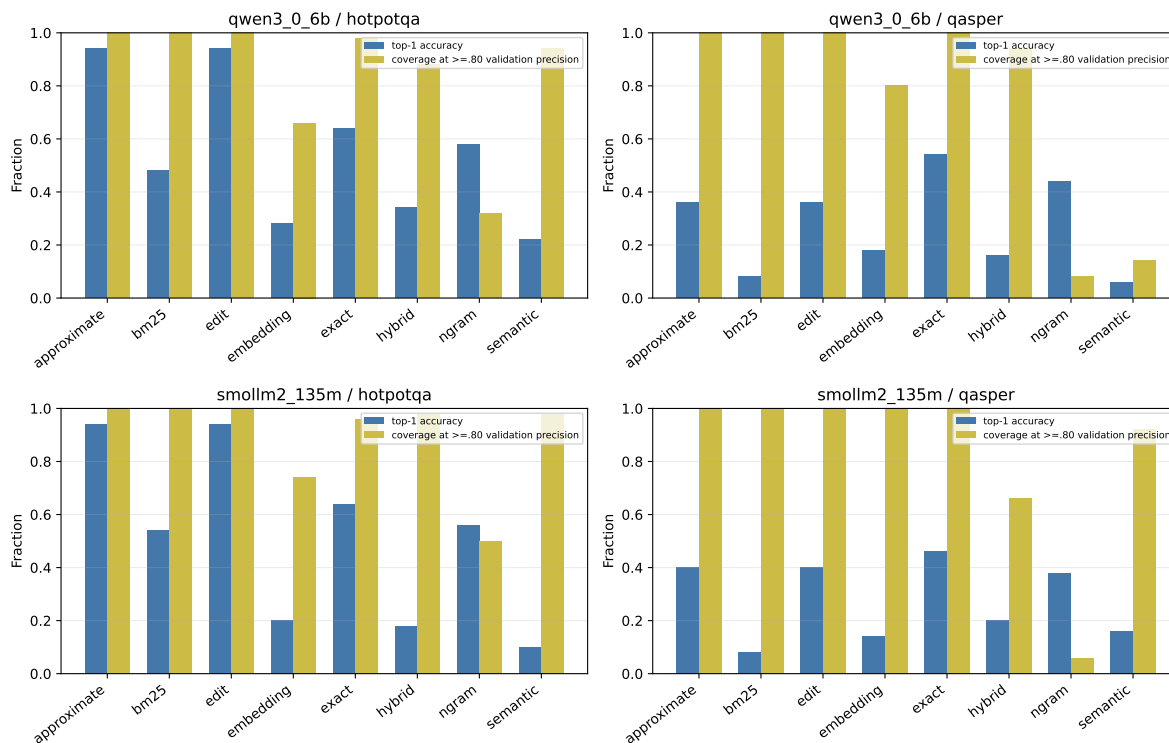


Figure 26: Natural top-reference correctness and retained coverage under validation-only thresholds. High coverage is not useful when held-out precision remains low.

14 Evaluation

Discovery metrics include supporting-document/fact recall, MRR, NDCG, precision, missed-evidence rate, hop success, and path completion. Resource metrics include discovered, requested, and materialized chunks/tokens, active fraction, distractor materialization, index construction, cold/warm routing, repeated-hop cost, transfer, and end-to-end latency.

14.1 Candidate-Normalized Discovery Efficiency

Absolute Top- K is not comparable across sources with different numbers of chunks or evidence spans. For each example, let N be the number of candidate chunks, E the number of evaluator-labelled evidence chunks, and K the deduplicated selected set. We report evidence recall $R_E = |K \cap E|/|E|$, evidence precision $P_E = |K \cap E|/|K|$, complete recovery $C_E = \mathbb{1}[E \subseteq K]$, selected fraction $\phi = K/N$, sparsity $\rho = 1 - \phi$, and working-set overhead $\omega = K/E$. For a successor route, K is the union of root and successor identities; thus duplicate selections do not receive extra credit. Search comparisons and token-span operations remain separate costs, not proxies for physical K/V activation.

Table 8: Held-out normalized efficiency at the fixed two-root plus two-successor budget. Each row is the highest-recall measured root–successor pair for that cohort. $C_E = 0$ throughout: the sparse policy improves partial recovery but is not a complete-evidence policy at this budget.

Cohort	Root $\rightarrow$ successor	R_E	P_E	K/N	K/E
QASPER	exact $\rightarrow$ exact-address	.121	.219	.032	1.029
HotpotQA	approximate $\rightarrow$ semantic	.145	.156	.088	.896
2Wiki	exact $\rightarrow$ semantic	.148	.236	.102	.901
MuSiQue	BM25 $\rightarrow$ exact-address	.086	.278	.050	.327

The normalization changes the interpretation in two useful ways. First, QASPER’s lower absolute recall occurs while touching only 3.2% of a much larger candidate pool; its best pair nevertheless selects about one chunk per evidence chunk. Second, MuSiQue’s best measured pair has the lowest working-set overhead but also the lowest recall. Neither comparison supports a universal channel: the Pareto point changes by cohort, in the same direction as the root/successor geometry analysis.

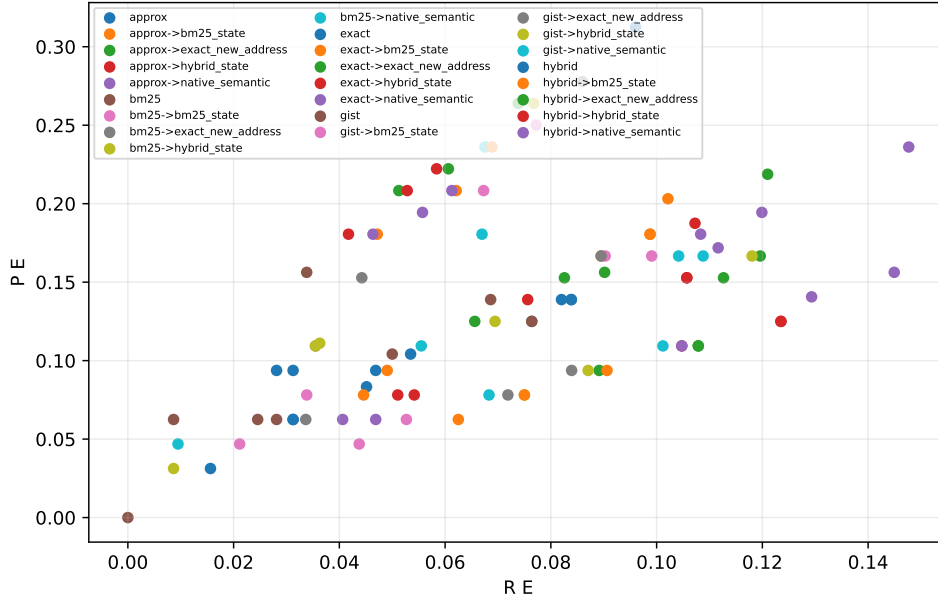


Figure 27: Evidence precision against evidence recall for every held-out root and root–successor policy. The full 25-pair surface is retained rather than only the cohort-wise maxima in Table 8.

A useful descriptive efficiency quantity is

$$E_R = \frac{\text{required evidence recall}}{\text{materialized token-KV}},$$

but Pareto curves are primary because a single ratio can hide quality–cost tradeoffs.

The normalized table above describes the original discovery-only cohort. The new Gate B confirmation follows the selected identities through native-K/V materialization and reports answer EM/F1, gold-answer log probability, timing, and physical memory. Confidence metrics include Brier score, expected calibration error, reliability curves, and selective accuracy/coverage. Disabled, shuffled, irrelevant, empty, and bounded direct-context controls are evaluated on the same held-out identities.

15 Open Ablations and Controls

This study covers the complete policy matrix, four natural discovery cohorts, matched primary budgets, raw and normalized exact spans, token n -grams, IDF and fixed-stop policies, bounded edit and host-embedding diagnostics, automatic aliases, two tokenizer/model families, per-example geometry, channel diversity, oracle headroom, a five-seed validation-trained selector, perturbations, natural and controlled wrong references, and provenance-aware calibration. The interaction experiment crosses channel with chunk size, chunk budget, and routing layer. It does not exhaust gist count, attention-layer count, models above 0.6B parameters, learned contextual entity linking, incremental index updates, or a substantially larger selector-training set. Those remain open rather than implied completed experiments.

Metadata must not leak gold supporting labels. The exact query text/state used by each channel and hop is logged. Discovery is not counted as materialization, and hybrid systems do not receive an unreported larger active budget.

16 Interface to Materialization

The implementation exposes the discovery-side equivalent of

$$\text{discover}(q, M, B, \pi) \rightarrow \{(r_i, \mathbf{s}_i, p_i, c_i)\}_{i=1}^n,$$

where $\mathbf{s}_i$ preserves channel scores, p_i provenance, and c_i has the referent-validity semantics defined above. Paper 3 consumes these candidates and independently chooses an active native-KV set:

$$\text{materialize}(C, B_{KV}, \pi_M) \rightarrow \mathcal{A}.$$

Paper 3 can consume this record without making materialization depend on one retriever.

17 Search-Method Action-Space Handoff

The version-2 action specification, `search_method_action_spec.json`, freezes five independent root actions:

$$S_{\text{root}} \in \{\text{semantic, exact, BM25, approximate, hybrid}\},$$

and five successor actions:

$$S_{\text{succ}} \in \{\text{native semantic, exact new address, BM25 state, approximate new address, hybrid state}\}.$$

Each entry names its implementation identifier, required state and indexes, legal parameters, confidence outputs, separate cost metrics, and known failure modes. Legacy channel names such as **gist** and **approx** remain explicit aliases rather than silently changing prior artifacts. The version-2 schema declares that root and successor choices are independent and records that its source experiment did not materialize K/V; the newer Gate B artifact records physical counts separately rather than rewriting that provenance.

Paper 2.6 establishes this search-method action space and its empirical heterogeneity. Paper 3.5 treats S_{root} and S_{succ} as adaptive actions jointly with query interpretation, breadth/depth, admission budgets, and bounded retry [9]. The handoff exports observable confidence and abstention signals, but it does not import dataset identity or gold outcomes into the controller.

18 Relation to the PRA Series

The position paper separates a large logical address space from the finite active working set and frames activation as demand-driven virtual memory [12]. Paper 2.6 studies discovery over that logical memory. Paper 1 establishes model-bounded native-K/V mechanics [13]; Paper 1.5 owns positional transport and relocation, which this study holds fixed. Paper 2 integrates frozen pretrained backbones and identifies routing, rather than K/V construction, as the limiting interface [14]; the present work changes discovery representations without retraining the consumer.

The strongest conceptual link is Paper 2.5’s associative-memory decomposition [10]. Query-to-root entry is often harder than memory-to-memory traversal, and admitted evidence can expose a new address. Paper 2.6 adds multiple retrieval representations for entering and traversing that topology. Paper 3 asks how discovered identities should be materialized at variable detail and budget [11]; Gate B holds one such contract fixed only to validate the discovery-to-consumption boundary. Paper 3.5 receives the method and confidence contract for adaptive selection. This paper does not claim that its diagnostic selector is that adaptive controller.

19 Related Work

19.1 Retrieval-Augmented Attention and Long-Context Memory

Dense passage retrieval and RAG select external text that a reader or generator subsequently consumes [3, 4]. RETRO retrieves neighboring text chunks into a trained decoder, while Memo-rizing Transformers query a non-differentiable memory of prior key/value representations [1, 18]. Query-aware sparse-attention systems operate closer to physical attention: Quest selects K/V pages from query-dependent bounds, and RetrievalAttention builds attention-aware vector indexes over K/V [15, 5]. Paper 2.6 is primarily upstream of that physical choice: it compares representations for discovering identity-bearing PRA memory objects. Gate B then holds the inherited Paper 2 materializer fixed to test whether discovery differences survive native-K/V admission; policies for choosing how much detail to materialize remain Paper 3’s subject.

19.2 RAG, Sparse Retrieval, and Static Hybrid Fusion

BM25 is a probabilistic sparse relevance model, whereas DPR learns dense passage similarity [8, 3]. Combining sparse and dense candidates, reranking them, or fusing ranks is established information-retrieval practice; reciprocal-rank fusion is one simple score-scale-independent method [2]. Hybrid IR itself is not the novelty here. The narrower question is whether exact, BM25, approximate, and hidden-state semantic representations occupy different regimes when they address identity-aligned

PRA memory. The negative static-fusion and RRF results show that complementary candidates can dilute a two-chunk budget; they do not imply that hybrid retrieval is generally ineffective.

19.3 Multi-Hop and State-Dependent Retrieval

Multi-hop dense retrieval conditions a later retrieval step on an earlier passage [19]. IRRR iteratively builds a reasoning path, generates the next search query, reranks evidence, and decides whether to continue [7]; IRCOT interleaves retrieval with generated reasoning [16]. These systems establish that retrieved evidence can change the next search state. PRA shares that state dependence but traverses identities aligned to reusable native-memory objects. Paper 2.6 further asks whether the representation itself should change between root entry and successor traversal.

19.4 Entity and Approximate Reference Resolution

Approximate string matching studies principled retrieval under insertions, deletions, substitutions, and related surface variation [6]. Entity linking addresses the harder problem of assigning a mention to the correct entity under ambiguity. BLINK, for example, retrieves entity descriptions with a bi-encoder and reranks mention–entity pairs with a cross-encoder [17]. The present exact and approximate channels are intentionally lighter-weight. Their typo and alias gains coexist with stale-alias, wrong-instance, and wrong-relation failures. Contextual entity disambiguation is therefore a relevant future component, not a capability claimed by the current matcher.

Table 9: Representative neighboring systems. “Activation memory” distinguishes Memorizing Transformers from layer-native PRA K/V; “exported” means Paper 2.6 defines an action for a later controller rather than selecting it here.

System	External text	Native K/V	Lexical	Semantic	Iterative/stateful	Adaptive channel
DPR / RAG	yes	no	no	dense	mostly one-shot	no
BM25+dense / RRF	yes	no	yes	yes	typically no	fixed fusion
MDR / IRRR	yes	no	query dependent	yes	yes	no
RETRO	yes	no	no	chunk neighbors	decoder conditioned	no
Memorizing Transformer	no	activation memory	no	key similarity	decode query	no
Quest / RetrievalAttention	no	yes	no	query–key	decode query	no
PRA Paper 2.6	logical refs	candidate native memory	yes	native gist	root/successor	exported

20 Contributions

This work contributes: (1) a provenance-preserving, multi-representational discovery formulation for PRA; (2) a model-token-native matcher evaluated beside decoded-word BM25; (3) posting-list indexes for exact spans, token n -grams, BM25, approximate narrowing, aliases, and explicit URIs; (4) one bounded iterative engine shared by semantic, token, and hybrid policies; (5) a four-dataset, matched-budget precision–recall and root/successor analysis; (6) a two-tokenizer invariance and index-scaling study; (7) a fresh two-model, 100-held-out-identity Gate B confirmation through native-K/V materialization and generation; (8) complete matched-budget causal controls and natural wrong-reference calibration; (9) leakage-audited five-seed selectors and oracle-headroom analysis; and (10) a versioned root/successor action space for adaptive control and a clean identity-level handoff to materialization.

21 Discussion

Established. PRA-native discovery is multi-representational and state dependent. Exact, BM25, approximate, and semantic channels each recover evidence missed by another representation; root and successor preferences differ on three of four original datasets; and neither static score fusion nor RRF dominates under the matched budget. Indexed token-native routing preserves exhaustive top-four results while sharply reducing scored candidates. Hotpot/QASPER channel geometry replicates across Qwen and SmolLM. Under frozen native-K/V consumption, Hotpot correct-memory likelihood gains are causal for both models; on QASPER, oracle-versus-irrelevant effects are resolved even though fixed-channel gains over disabled memory are not.

Unresolved. Observable method selection, referential disambiguation, and routing-to-consumption transfer remain unsolved. Five controller seeds are stable but all remain below the best fixed channel. Validation-CDF selectivity does not transfer except for approximate Hotpot, and QASPER’s high retrieval recall does not produce a fixed-channel likelihood gain. Contextual entity/relation resolution, stronger consumers, and unmodified full-document cohorts are needed before an adaptive policy can be trusted.

Handed off. Paper 3.5 receives independent root and successor actions, observable confidence fields, failure indicators, and abstention opportunities. Paper 3 receives selected conceptual identities and independently controls physical K/V admission. Gate B measures one fixed materialization contract; it does not collapse discovery and materialization into one policy.

22 Limitations and Falsification

The original frozen natural study contains 132 identities, of which 74 are held out: 16 QASPER, 16 HotpotQA, 24 2Wiki, and 18 MuSiQue. Only 18 held-out 2Wiki examples have a distinct annotated successor under this partition. Its inherited feature cache remains one-model and cannot be enlarged without recapture. The confirmation adds 40 disjoint validation and 100 disjoint test identities, evaluated under both model families; the interaction cohort adds 16 more. Five seeds measure optimization sensitivity, not independent dataset samples.

The confirmation’s evidence-first 1,536-token construction depends on gold spans to retain evidence and is therefore a mechanism experiment, not an unmodified QA benchmark. Only Qwen3-0.6B and SmolLM2-135M are tested; neither is a strong QA generator, and EM/F1 stays weak. The interaction crossing uses eight examples per dataset and is exploratory. QASPER and Hotpot successor labels are unordered evidence remainders rather than annotated dependency edges. Geometry strata are descriptive and partly gold-derived; they explain failures but cannot be controller inputs. Natural wrong-reference labels indicate evidence overlap rather than adjudicated entity-coreference correctness.

The index memory number is a payload estimate that excludes Python-object overhead. Updates rebuild full corpus statistics rather than append incrementally. Exact lookup is much faster than exhaustive scoring, but the 0.50–0.65 s fuzzy paths at 4,096 chunks remain too slow for serving claims. Validation-calibrated confidence does not transfer reliably, neither selector closes the oracle gap, and confidently wrong or stale references remain possible.

The end-to-end result is deliberately metric specific. Hotpot gold-answer likelihood improves causally, but generated EM/F1 does not. QASPER correct memory beats irrelevant memory at the oracle ceiling, yet no fixed channel beats disabled memory with a resolved interval. A stronger

claim would require gains under unmodified documents, larger models, fresh datasets, and answer-quality metrics. Failure to reproduce indexed parity, cross-tokenizer channel ordering, or the paired correct-versus-wrong-memory effects would falsify the corresponding claims independently.

23 Conclusion

PRA discovery need not be one hidden-state similarity function. An identity-aligned token sidecar can participate in bounded gist traversal, use posting lists rather than exhaustive candidate scans, and pass selected identities into native-K/V materialization without changing the frozen consumer. Across four discovery cohorts, exact, BM25, approximate, and semantic channels occupy different regimes, and the preferred representation changes from root entry to successor traversal on three datasets. Across two fresh model/tokenizer families, BM25 again leads Hotpot retrieval and approximate matching leads QASPER.

The supported conclusion is architectural and operationally bounded. Retrieval representation should be exposed as a state-dependent action, with S_{root} separate from S_{succ} . Hotpot shows that correctly discovered sparse memory can improve frozen-model answer likelihood after physical admission; QASPER shows why discovery recall cannot substitute for consumption evaluation. Until observable selectors, referential validity, and generated answer quality improve on larger unmodified cohorts, dataset-validated fixed policies, explicit references, abstention, and bounded retry are more defensible than a universal fusion rule.

References

- [1] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George B. M. van den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, et al. Improving language models by retrieving from trillions of tokens. In *International Conference on Machine Learning*, 2022.
- [2] Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759, 2009.
- [3] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of EMNLP*, 2020.
- [4] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, 2020.
- [5] Di Liu, Meng Chen, Baotong Lu, Huiqiang Jiang, Zhenhua Han, Qianxi Zhang, Qi Chen, Chengrui Zhang, Bailu Ding, Kai Zhang, Chen Chen, Fan Yang, Yuqing Yang, and Lili Qiu. Retrievalattention: Accelerating long-context LLM inference via vector retrieval. In *Advances in Neural Information Processing Systems*, 2025.
- [6] Gonzalo Navarro. A guided tour to approximate string matching. *ACM Computing Surveys*, 33(1):31–88, 2001.

- [7] Peng Qi, Haejun Lee, Tg Sido, and Christopher Manning. Answering open-domain questions of varying reasoning steps from text. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 3599–3614, 2021.
- [8] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009.
- [9] Jorge Simao. Adaptive progressive retrieval attention: Interpretation, search-method selection, admission, and bounded retry. Manuscript, 2026.
- [10] Jorge Simao. Associative memory in transformers: Topology, iterative retrieval, and controlled activation with progressive retrieval attention. Manuscript, 2026.
- [11] Jorge Simao. Computationally sufficient memory: Causal native-k/v materialization for progressive retrieval attention. Manuscript, 2026.
- [12] Jorge Simao. Position, attention geometry, and semantic routing for retrieved native-KV memory. Companion Paper 1.5 manuscript, 2026.
- [13] Jorge Simao. Progressive retrieval attention: Bounded sparse native-KV for addressable logical memory. Companion Paper 1 manuscript, 2026.
- [14] Jorge Simao. Progressive retrieval attention for pretrained transformers: Sparse native-K/V memory with semantic routing. Companion Paper 2 manuscript, 2026.
- [15] Jiaming Tang, Yilong Zhao, Kan Zhu, Guangxuan Xiao, Baris Kasikci, and Song Han. QUEST: Query-aware sparsity for efficient long-context LLM inference. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.
- [16] Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 10014–10037, 2023.
- [17] Ledell Wu, Fabio Petroni, Martin Josifoski, Sebastian Riedel, and Luke Zettlemoyer. Scalable zero-shot entity linking with dense entity retrieval. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 6397–6407, 2020.
- [18] Yuhuai Wu, Markus N. Rabe, DeLesley Hutchins, and Christian Szegedy. Memorizing transformers. In *International Conference on Learning Representations*, 2022.
- [19] Wenhan Xiong, Xiang Lorraine Li, Srinivasan Iyer, Jingfei Du, Patrick Lewis, William Yang Wang, Yashar Mehdad, Wen-tau Yih, Sebastian Riedel, Douwe Kiela, and Barlas Oguz. Answering complex open-domain questions with multi-hop dense retrieval. In *International Conference on Learning Representations*, 2021.